



Machine Learning in Stock Return Prediction: Evidence from In-Sample and Out-Of-Sample Tests

How Machine Learning Practices Perform in Rolling and Expanding window Approaches

Master Thesis Finance

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Abstract

The goal of this study is to evaluate whether machine learning and regularized regression models are able to provide any improvement in stock return predictions compared to conventional linear regression models for real world use. Using monthly data from the United States from 1941 to 2024, we evaluated Lasso, Ridge, XGBoost and Random Forests against OLS in-sample, expanding windows, and rolling windows. Although the ML models performed well in-sample, their predictive ability from the out-of-sample data collapsed. The error metrics of each model remained stable from in-sample to out-of-sample results, validating the challenge of predicting returns. Returning to the goals of practical usage and predictive improvement with regard to the complexity of models, the present study demonstrates caution when drawing conclusions about the predictive abilities of sophisticated models in practice. Our findings underscore additional considerations for predicting financial values, including scrutiny of out-of-sample validation and hybrid approaches such as the combinations of algorithms, models, and data.

Introduction

Predicting stock returns is one of the most important and most studied issues in financial research for both academic and applied researchers, practitioners, and regulators. When predicting stock returns accurately, it has several implications for asset pricing models, portfolio management, and the management of risk. Despite decades of substantial empirical research, the question of whether or not stock returns are predictable remains an open question. In fact traditional financial models have been criticized for being based on traditional linear regression, while recent developments around machine learning (ML) now include non-linear modeling approaches that benefit from can account of both the variability of combinations and the possibility of controlling for overfitting using principles of regularization and ensemble modelling. However, we currently have important gaps of knowledge about the strength of these modelling approaches we the economic conditions, along with establishing the extent to allow us to make valid conclusions about which predictors were capable of maintaining their predictive value through various states of recovery, decline expansion, and contraction.

This thesis will compare of modelling approaches (Lasso regression, Ridge regression, XGBoost and Random Forests) to predict stock returns using twelve potential predictors that incorporate macroeconomic and firm-specific predictions. For the analysis, OLS linear regression will provide a contextual foundation, with two aspects fulfilling the required role of providing context: OLS linear regression can verify theoretically plausible linear relationships between predictors and returns, and it serves as a foundational model for contrasting more advanced methods. The predictors selected in this study on stock return prediction are all utilized regularly in the finance literature on asset pricing, including ratios of valuation (dividend-price, price-earnings), macroeconomic parameters (CPI, bonds long term yield), financial spreads (term difference, default return spread) that provide the market information that incorporates fundamental performance, economic conditions, and risk appetite. Potentially less studied variables such as net equity issuance and the default return spread have been added to assess their performance and the consistency across models.

To evaluate the performance of each model, the authors employ three different testing paradigms: in-sample, expanding window out-of-sample (OOS) prediction, and rolling window OOS. The in-sample test provides an "apples-to-apples" evaluation, whereas both expanding and rolling window OOS models intended to emulate investment behaviour, the goal of which is to update the model periodically to gain value from information. The authors employ the expanding window approach, which continually expands the analysis with new observations in a temporal sequence (i.e. gradually gaining a larger base of knowledge) and the rolling window, which maintains a consistent temporal horizon but prioritizes more recent data and adapts to changing market conditions. Importantly, the OLS linear regression benchmark is applied consistently across all testing environments, ensuring that any improvement (or not) that can be attributed to Lasso, Ridge or machine learning-based methods come from their ability, and not a trait or misspecifications in the data or models.

Employing a rich set of monthly data; from April 1941 to December 2024- over multiple economic cycles, we applied a longitudinal modeling framework that considered the complexity of having different macroeconomic conditions occurring over a more than 80 year horizon. The out-of-sample forecast is derived from a 40 year period (April 1941-April

1981) before forecasting for the next decades of uncertain economic conditions. The retrospective forecasting allows for an increased degree of statistical rigor as it takes into account the possible serial correlation, and examines whether forecast relationships will hold up over time as economic conditions change.

The study's overarching objective; in exploring three key comparisons of penalized regression (Lasso & Ridge), ensemble-based machine learning (XGBoost, RF), and, as the benchmark, an OLS linear regression model: Firstly, do machine learning models or regularized regressions consistently outperform OLS linear regression models either for in-sample or out-of-sample stock return predictions? Secondly, how sensitive are the models to the evaluation window- should I be using an expanding or rolling window, and which method produces more stable forecast? Thirdly, which predictors are consistently stable and significant across time horizons and modelling method?

The suggestions from the study will add both to the theoretical academic literature, and to the practical managers of stock portfolios. In theory, the results contribute to an understanding of the increased utility of contemporary machine-learning methods over classical econometric models in understanding the complexities associated with financial datasets while also acknowledging the role that linear regression continues to play in diagnostics and benchmarking. In practice, the findings may provide some pathway for managers and quantitative analyst to identify ways to model alternative frameworks that have both the best combination of accuracy and computational efficiency. Likewise, the finding in relation to the comparative stability of the predictor variables between the modelling methods may further inform some existing frameworks around an asset pricing question, and may help provide some insight around an emerging set of new financial forecasting modelling methods available to regulators and analyst.

Literature Review

Typical Considerations and the Efficient Market Hypothesis

Literature on stock return predictability is extensively grounded in the Efficient Market Hypothesis (EMH), introduced by Fama (1970). The EMH states that all available information is reflected in asset prices and that there is no reason to believe that one can make excess

returns systematically. Fama's argument that stock prices follow a random walk discourages any attempts to forecast stock returns. However, subsequent empirical research has identified anomalies that suggest even the stringent interpretations of the EMH are incorrect. For example, Rapach et al. (2010) showed that including macroeconomic variables (e.g., interest rates, inflation) into linear predictive models improves forecasting power, particularly in out of sample test. The study also showed that when predictors are combined, it helps alleviate model instability, suggesting that even basic linear methods can support forecasting even if time series data face structural breaks. In a bid to improve prediction methods, Campbell and Thompson (2008) applied economically meaningful restrictions (e.g. positive equity premia) to the regressions and found that these restrictions generally improved out-of-sample performance. In contrast, Welch and Goyal (2008) looked at a more extensive list of predictors (e.g., dividend yield, term spread), finding that most of them did not significantly improve upon historical mean forecasts, but with caveats about certain market conditions. Taken together, these studies provide information about the limitations, as well as the potential for linear methods, while pointing out how critical the predictors' selections are. Our discussion also lays the groundwork for moving towards more complicated approaches, such as machine learning, which accounts for the non-linearity and the potential complexity of financial data.

Machine Learning in Finance

Over the last few years, advances in machine learning have transformed return prediction by successfully accommodating significant hurdles such as non-linearity, interaction effects, and high-dimensional data. Gu et al. (2020) conducted a comparison of a range of machine learning algorithms and classical econometrics models. Hong et al. 2018 demonstrated that the tree models consistently outperformed the linear models in both in-sample and out-of-sample tests. Fischer and Krauss (2018) similarly found that XGBoost outperformed classical logistic regression, and deep learning models in predicting return directions for S&P 500 stocks because of its ability to model complex nonlinear relationships between features. The benefits of ML in modeling structures are clear, especially in turbulent environments. Patel et al. (2015) studied the performance of Random Forests on forecasting volatility; the Random Forests model outperformed ARIMA and GARCH models, and in a turbulent environment, it may have outperformed a combination of ARIMA and GARCH models that

were undoubtedly influenced by outliers (cf., financial crises). Lim and Zohren (2021) showed that XGBoost was better than traditional time series models in a high dimensional setting, while also outlining the limitations of traditional time series models, such as model resolution and ability to regularize when examining text data from earnings calling transcripts. However, Butaru et al. (2016) noted that ML models can over-fit when signal is low, which reinforces our choice to concentrate our efforts on out of sample and the priori consideration while designing this study.

Regularized Regressions

In the ML world, classical regularized (e.g., Tree-based) regressions are still relevant. For instance, the Lasso regression from Tibshirani (1996) improves prediction by modifying regression coefficients and setting some to zero, thereby fixing overfitting and not explicitly selecting variables. Hastie et al. (2009) prescribed Ridge regression as a potential remedy to multicollinearity among predictors (e.g., value ratios), while De Mol et al. (2008) said that the Lasso worked very well under conditions of sparsity in the data. But several issues emerged with Lasso, as Rapach and Zhou (2020) acknowledged that Lasso appears to provide weak recommendations when variables were highly correlated (e.g., interest rate variables) at all forecast horizons, and rather endorsed a type of hybrid like Elastic Net. This prior research supports the notion that using some type of combination of methods beside their own, is often beneficial. For instance, in their meta-analysis, Rapach et al. (2013) evidenced using forecast combinations (mean, median, trimmed) to the inclusion of multi-period forecasts, increased the overall reduction in variance. Along the same lines, Kim et al (2021) showed that combining XGBoost forecasts selected through Lasso led to better risk-adjusted returns. Finally, Wang et al. (2022) proposed a two-stage approach which extracted and forecasted interval Random Forests from selected Lasso variables produced better forecasts than just using either Lasso or Random Forests alone in forecasting multi-period S&P 500 Returns.

Market Regimes and frequencies

Predictive performance reflects a record of market regimes. For example, Henkel et al. (2011) showed that dividend yields predicted returns when the economy was in recession. Dangl and Halling (2012) claimed that rules could better capture regime switching by using

moving and rolling-window regressions, and noted if their specification uses an expanding-window estimate, that it is able to capture the impact of a regime-shift. Borovička and Hansen (2014) referred specifically to low-frequency data in developing for the intramonthly effects when using the ML methods (Random Forests) that produced superior results to using fundamentals approaches. In conclusion, there is consistent evidence for both the necessity for adaptive delivery of models for evolving scenarios and markets, particularly during turbulent or illiquid conditions. As an example, Özdemir and Güngör (2021) based a hierarchical regional call on the features of Turkey's BIST 100 as an emerging market - in terms of relatively high volatility could arguably have at least several distinctive features that require an appropriate regional call and/or model.

Critiques, Gaps and Future Questions

While there is a certain optimism regarding ML, critics raised some degree of caution. Chincó et al. (2019) noted 'p hacking' in high dimensional ML, asserted that even more stringent significance cutoffs need to be used. Bryzgalova et al. (2023) explained that many of the ML based anomalies were not 'out of sample' replicable, further demonstrating the systematic development of alternative procedures was required to test for out of sample research rigorously and ensure reproducible and robust results in Subsequent sampling (e.g., 1941-2024) achieved the same results of reproducible results. These criticisms suggested that future research projects should take a look at what dynamic frameworks look like but that are updated while practitioners are unaware of the changes.

Data Sources and Construction

1. Dividend-to-Price Ratio d/p :

The dividend-to-price ratio (d/p) was created with monthly data from Shiller's historical data set (1871 - present). This data set is consistent; all 12 monthly data points are contiguous and there are no missing values or marked structural breaks.

1.1. 12-Month Moving Sum of Dividends (D12):

- The dividend was totalled for the previous 12 months

1.2. Logarithmic Transformation:

- The dividend - price ratio was calculated as the following equation:

$$d/p = \log (D_{12}) - \log (P_t)$$

Where P_t is the stock price at time t .

1.3 Multicollinearity Problems and Adjustments

Before modelling, all-time series were analysed for stationarity as a diagnostic check, utilizing the Augmented Dickey-Fuller (ADF) test. The ADF results indicated the raw dividend-price (d/p) series had characteristics of unit root and thus were non-stationary. This is theoretically plausible, given that trends in long-term price levels may create spurious correlations in regression, if neglected.

To address this limitation, while demonstrating bindings with cyclical components, I ran the Hodrick-Prescott (HP) filter with a smoothing parameter of $\lambda=14,400$, which is a benchmark for monthly data (Ravn & Uhlig, 2002). The HP filter decomposes the series into trend and cyclical components, and only cyclical components will be treated as a covariate in the remaining regressions. Consequently, in assessing regression assumptions (primarily stationarity), we can achieve meaningful cyclical fluctuations using the HP methodology, although they may not represent economic fluctuations.

Moreover, using the Hodrick-Prescott method that is widely published as such, and is employed for limiting spurious effects of regression (Hamilton, 2018). Our choice of $\lambda=14,400$ conforms with monthly data (Backus & Kehoe, 1992) to ensure we are consistent with the literature and helping to limit arbitrariness in the parameterization.

2. Earnings-to-Price Ratio (e/p):

The e/p is the earnings-to-price ratio derived from the S&P 500 Price-to-Earnings (P/E) ratio from Macrotrends 90-Year Historical chart (1926 - now). Although this was a shorter historical time-series for our common view when compared against Shiller's data, as it starts in 1926

2.1. P/E conversion to e/p :

$$e/p = 1 / (p/e)$$

This is consistent with the more traditional financial literature (e.g., Campbell & Shiller, 1988) which allows some comparability for predictive regressions.

2.2 Multicollinearity Problems and Adjustments

Using the same filtering technique as before for the dividend-price (d/p) ratio—the Hodrick-Prescott (HP) filter ($\lambda = 14,400$) to preserve the cyclical component of the earnings-price (e/p) ratio (i.e., e/pcycle)—and running the regression with both e/pcycle and d/pcycle, we found severe multicollinearity, with Variance Inflation Factors (VIF) exceeding 5. Correlation analysis performed on the cyclical components indicated that were strongly correlated (Pearson $r \approx 0.65$). This would be expected since they are responding to the same fundamental trends in corporate earnings and dividends. In this context, we had two possible options to rectify the multicollinearity issues. We considered, dropping either one of the filtered variables or utilizing the unfiltered e/p ratio specification which is the 'standard' specification for return prediction studies (Fama & French, 1988). We chose the second road to bypass adding arbitrary cyclical signals while sustaining consistency with the established empirical literature. This decision also ensures unbiased estimation of OLS standard errors (Gujarati, 2009).

I have performed diagnostic checks to confirm I had modeled correctly. First, the unit root tests established onto earnings price (e/p) ratio series was stationary (Augmented Dickey-Fuller test) so detrending was not needed. Second, by the time we settled on our predictor specifications, the variance inflation factors (VIFs) for each predictor were less than 2, so there is no concern for multicollinearity, and based on our variable selection process we had already ruled out multicollinearity issues. The various robustness checks that we conducted support the validity of our specification and confirm that the results are statistically valid.

3. Dividend Payout Ratio (de):

Each calendar month, the dividend payout ratio (de) was calculated utilizing Shiller's historical dataset (1871-present) to maintain cohesion between the other variables of interest in the study. The full standard form of the logarithmic specification was as the following:

$$d/e = \log(D_{12}) - \log(E_t)$$

Where: D_{12} : the 12-month moving sum of dividends (the same series used for d/p)

E: Earnings for month t

3.1 Multicollinearity Problems and Adjustments

To investigate the time-series properties of the d/e ratio, we first implemented an Augmented Dickey-Fuller test which revealed that the raw series was stationary. The d/e series had no evidence of trending behaviour, though, unlike the d/p ratio, did show some stationary behaviour without any need for detrending.

There is some theoretical justification for the raw, specification as ratio-type variables (i.e. the d/e ratio) are sure to diminish long-term trends, as well as have cyclical variation in conjunction with business conditions (Cochrane, 2011). Using raw series allows researchers to utilize the most amount of available data, with the assurance that earning changes consistently track-quality shifts in the economic cycle and expectation.

Finally, we investigated multicollinearity in the full model with variance inflation factors (VIFs). The d/e ratio had only very limited collinearity ($VIF = 1.29$), as we might expect, so was a generally reliable independent predictor.

4. Stock Return

Stock market returns, produced by using a few simple financial metrics, are the primary dependent variable for this study. In the WRDS platform's CRSP database, we extracted the equally-weighted return series (*ewretd*) starting January 1926 and were meticulous in our choice. The *ewretd* time series is backed by depth in the database as well as a holistic representation of today's market, it takes an equal contribution from each of NYSE, AMEX and NASDAQ in calculating this monthly return.

The construction method is also crucial. The returns are representative of the real economic experience of a diversified investor who fully participates in capital appreciation and income production - reinvesting dividends and relevant corporate actions. Conceptually, the treatment follows the gold standard of fundamental asset pricing research (Fama & French, 1993), and avoids survivorship bias from many commercial indices by counting firms/delisted firms.

The fact that this return series is simply constructed, or not transformed in any way, also lends itself to prediction modelling. These returns, which are percentages, provide similar economic value over nearly a century of market history - from the Great Depression to the Digital Age. Our temporal span allows us the statistical power to examine trivial but persistent return patterns and time-spectral-trends that shorter periods would not allow.

Beware of value-weighted returns versus equal-weighted returns. The equal classification of firms is irrespective of their ultimate purpose or goal. Value-weighted indices represent mega-cap stock price movements disproportionately i.e. with the "Big" 5 tech stocks dominating price returns, our measure offers firms of all sizes smaller and larger proportions in the returns. This is important for researching return predictability as Banz (1981) demonstrated the small-firm effect whereby smaller firms do have different risk-return characteristics than larger firms hence our present motive - equal-weighting all firms gives us a cleaner picture of average listed firms response to average market dynamics like dividend policy, and earnings expectations.

5.Consumer Price Index

The Consumer Price Index (CPI) was obtained from Shiller's long-run dataset (1871 – Present) and provided a consistent monthly measure of inflation. The raw CPI has an upward trend over time because the price level is increasing over time, so we transformed it to be stationary:

$$\Delta \log (CPI_t) = \log(CPI_t) - \log (CPI_{t-1})$$

This cannot be a model of the raw price index monthly inflation since they have a long-term trend, it is a transformation taking the raw price index and turning it into monthly inflation rates. Using this transformation will effectively remove the long-term trend away from the monthly price levels but keep the short-term price level variance. The resulting series provides a clean and interpretable measure of inflation that is widely accepted in both financial and macroeconomic literature (Campbell & Vuolteenaho, 2004). The log-differenced series may also be used in a regression, and while the raw CPI had an upward trend, the log-differenced series does not contain drift or long-term drift, thus it is not vulnerable to spurious regression due to non-stationarity.

6. Long Term Yield

The 10-year Treasury yield (GS10) was taken directly from a monthly, continuous collection of long-term government bond yields previously downloaded from Shiller's database. There were two major adjustments made to create the variable and make it validate for time series analysis:

First, I rescaled the yield, by taking it from a percentage to decimal by dividing by 100 (e.g. 5.00% became 0.05) in line with financial modeling. After examining the data initially, I saw that there were persistent trends in the nominal yield series, a typical outcome in interest rate data where the long time-series presents both upward and downward trends.

In order to manage this non-stationarity, but retain the economic meaning of the yield changes, I applied a log-difference transformation to the yields:

$$\Delta \log (GS10_t) = \log(GS10_t) - \log (GS10_{t-1})$$

The transformed GS10 series will have the important characteristics of the long-term interest rate movement, plus it will be "correct" for modeling purposes, since it allows for nonstationarity. This is typical in financial econometrics, where log-difference transforms of rates are common (Ang & Piazzesi, 2003).

7. Long term government Bond Returns

The 7-year government bond returns came from WRDS (Wharton Research Data Services) using the CRSP Treasury database for historical account which satisfies a sufficiently long-time length for reasoning purposes without sacrificing fidelity to traditional fixed income research conventions. The raw return series was incorporated as is in earnings-return percentages placing emphasis on economic meaning.

It is near the intermediate point on the yield curve that captures fixed rate interest rate risk exposure to equity investors whilst steering clear of the long term bond's volatility.

8. Term Spread

The term spread (tms) was defined as the difference between long-term government bond yields (GS10) [for example the yield on a 10-Year bond] and 3-month Treasury bill rates

(TB3MS) representing the slope of the yield curve which is an important predictor of economic conditions and future stock returns.

The long-term yields were taken from Shiller's data set in original percentage, with the measure divided by 100 so they could be consistent with TB3MS. Long-term T-bill rates (short term) were obtained from FRED (Federal Reserve Economic Data) through R's quantmod package (the series is `r getSymbols("TB3MS", src = "FRED")`), with the monthly series going back to January 1934 and similarly converted to decimal by dividing by 100.

tms was finalized simply by plugging in the numbers as $tms = GS10 - TB3MS$. This measures the yield premium for an investor holding a long-term government bond vs a short-term risk-free asset.

We convert tms into a mean-reverting series that is stationary by nature and suitable for regression analyses without additional detrending, and it retains its meaning for economic interpretations as a notable financial measure in much of the literature for return predictability and business cycle forecasting (Campbell & Shiller, 1991; Estrella & Mishkin, 1998).

9.Default Yield Spread

The default spread (dfy) was calculated as the yield spread between the BAA-rate and AAA-rated corporate bonds and reflects the risk premium that investors require for holding lower-grade corporate debt in lieu of investment-grade bonds.

The yield data for both series was simply raw yield data collected monthly from FRED (Federal Reserve Economic Data) with the use of R's quantmod package with the command `getSymbols("AAA", src="FRED")` for AAA-rated corporate bond and `getSymbols("BAA", src="FRED")` for BAA-rated corporate bonds, which both went back several decades allowing for enough historical coverage.

The final default spread measure was calculated as $dfy = BAA - AAA$, which is the yield add-on for taking on higher credit risk in the corporate bond market since widening spreads are generally interpreted as increasing risk aversion and worsening economy while the compression of spreads either indicates improving credit conditions or increasing investor confidence.

10. Default Return Spread

The default return spread (dfr) was defined as the difference between returns on long-term corporate bonds (BAA-b7ret) and the long term government bonds, representing the excess compensation investors require for taking credit risk in the corporate market over risk-free sovereign debt.

The corporate bond returns were collected from FRED using the same process as above (BAA from FRED while calculating dfy), while the government bond returns were taken from the previously described Treasury series (ltr(b7ret)).

After conducting some initial analysis, we found that the spread time series had an underlying trend component, so we applied the Hodrick-Prescott filter ($\lambda=14,400$, the normal value for monthly data), in order to exclude the trend component from the time series while still preserving the economic relationship between the corporate and government debt markets.

The hodrick-prescott filtering produced a stationary time series that preserves the relevant credit risk information while being able to discard the potentially spurious effects of long-term trends, while still producing a measure of a time-varying risk premium that has been shown to predict future economic activity (Keim & Stambaugh, 1986) and equity returns (Gilchrist & Zakrajšek, 2012) in the finance literature.

11. Book to market Ratio

The book-to-market ratio (b/m) created unique data challenges, which required a multi-source approach to develop a complete historical series. When the sought after Value Line Long-Term Perspective Chart became unavailable to us, we turned to WRDS's Financial Ratios Industry Level database (Fama-French 49 Industries) with coverage starting in 1970.

For the period prior from June 1950 to December 1969, we obtained book value of equity (bkvlps) and shares outstanding (csho) values from CRSP/Compustat Merged - Fundamentals Annual and multiplied the components to yield total book equity, removing any NA values and then averaging by duplicate dates. The final book equity number was divided by each corresponding market value found from Shiller's price series to yield the b/m calculation.^[5]

To fill the period in the 110 months gap we used an ARIMA forecasting approach, we backcast 110 months from the existing series using the **auto.arima** function in R with the monthly count as the time variable in the series to ensure continuity with our data. This entire methodology resulted in a complete monthly book to market series within the constraints of established financial research, Fama and French (1992).

12.Net Equity Expansion (Ntis)

The net equity expansion is a metric measures total corporate financing activity by examining net stock issuance as a proportion of market conditions. We utilize CRSP's NYSE index data to remain consistent with the corporation finance literature and data at hand.

12.1 Construction Methodologies:

The initial calculation is net dollar issuance determined from current and lagged NYSE market capitalization adjusting for pricing changes:

$$NetIssue_t = Mcap_t - Mcap_{t-1} * (1 + vwretx_t)$$

where Mcap is the total NYSE market capitalization and vwretx is the value-weighted return net of dividends. This isolates the net dollar activity of equity financing activities when adjusting for valuations.

13.Stock Variance (Svar)

The stock variance measure was derived using daily S&P 500 returns pulled from the Yahoo Finance API through the quantmod package in R. The process began by calculating daily logarithmic returns:

$$r_t = \log(P_t) - \log(P_{t-t})$$

where P_t is the daily closing price of the S&P 500 index at time t. These daily returns were squared to make a variance proxy:

$$r_t^2 = (\log(P_t) - \log(P_{t-t}))^2$$

Monthly realized variance was calculated by summing these squared daily returns:

$$\sigma_m^2 = \sum_{t=1}^N r_t^2$$

where σ_m^2 represents a monthly variance estimate and N is the number of trading days in the month. It was determined in the original run of the analysis that contemporaneous variance was prognostically overfitting returns so we put it through a one-month lag structure (utilizing σ_{m-1}^2). This better attenuates the lagged relationship between one month of volatility and subsequent returns. This way, we follow conventional procedures put forth in the volatility forecasting literature (Andersen et al., 2003) and using summed squared returns as a variance proxy follows the realized volatility approach (Bollerslev et al., 2016). The result is a clear monthly measure of equity market risk that contains full information about volatility during the month but correctly timestamps (provides temporal structure) both volatility and returns.

A. Outputs And Methodology

A.1 Model specification

To evaluate the plausibility of the data and to exploratory the initial relationship between the predictors and the dependent variable, a linear regression model was estimated using ordinary least squares (OLS). For the model, stock returns (the dependent variable) were predicted on a set of predictors that were categorized by valuation ratios, macroeconomic variables and financial risk. The full specification of the model is:

$$\begin{aligned} stock_{return_t} = & \beta_0 + \beta_1 dp_{cycle_t} + \beta_2 ep_t + \beta_3 de_t + \beta_4 CPI_t + \beta_5 GS10_t + \beta_6 tms_t + \beta_7 dfy_t \\ & + \beta_8 dfr_{cycle_t} + \beta_9 b_m_t + \beta_{10} Ntis_t + \beta_{11} b7ret_t + \beta_{12} lagged_svar_t + \epsilon_t \end{aligned}$$

Where:

$stock_return_t$ represents the stock return at time t ,

dp_cycle_t detrended dividend-price ratio,

ep_t denotes the earnings-price ratio,

de_t is the dividend-earnings ratio,

CPI_t is the consumer Price index (inflation measure),

$GS10$ is the 10-year Treasury yield,

tms_t is the term spread,

dfy_t is the default yield spread,

dfr_cycle_t is the de-trended default return,

b_m_t is the book-to-market ratio,

$Ntis_t$ is the net equity issuance,

$b7ret_t$ is the 7-year government bond return,

$lagged_svar_t$ is the lagged stock variance

ϵ_t is the error term.

The model estimated in R using `lm()` function.

Table A.1 Result of the linear regression

Variable	Estimate	Std. Error	t-value	p-value	Significance
(Intercept)	-7.395e-03	1.106e-02	-0.669	0.50384	
dp_cycle	-3.833e-03	2.327e-02	-0.165	0.86921	
ep	-2.719e-02	5.612e-02	-0.484	0.62820	
de	4.083e-03	5.723e-03	0.713	0.47576	
CPI	-7.878e-01	3.603e-01	-2.187	0.02900	*
GS10	4.654e-02	3.645e-02	1.277	0.20198	
tms	6.509e-02	1.474e-01	0.442	0.65877	
dfy	1.178e-02	4.347e-03	2.710	0.00684	**
dfr_cycle	-1.232e-02	4.003e-03	-3.078	0.00214	**
b_m	3.242e-03	1.796e-03	1.805	0.07134	.
Ntis	2.662e-12	5.654e-11	0.047	0.96246	
b7ret	2.758e-01	1.117e-01	2.468	0.01374	*
lagged_svar	-2.459e-01	3.714e-01	-0.662	0.50804	

Model Statistics:

- **Residuals:** Min = -0.267; 1Q = -0.028; Median = 0.002; 3Q = 0.028; Max = 0.220

- **Residual Std. Error:** 0.04791 (df = 992)
- **R²:** 0.03247 , **Adjusted R²:** 0.02077
- **F-statistic:** 2.775 (df = 12, 992), **p-value:** 0.0009984

Significance codes:

*** p < 0.001, ** p < 0.01, * p < 0.05, . p < 0.1

A.2 Regression and Interpretation of Results

The first linear regression model of stock returns to a total of twelve predictors- valuation ratios, macroeconomic indicators, and financial risk factors- was found to be statistically significant but provided relatively weak explanatory power. The F-statistic (2.775, p = 0.000998) indicates that the statistical significance means that Stock returns can be explained by the predictors collectively, but the adjusted R² statistic for goodness-of-fit (0.0208) indicates that predictors explain only a limited amount (2.08%) of the variation in returns. This reinforces previous literature indicating that macroeconomic and financial predictors do not demonstrate short-term predictability as a result of market efficiency or unobserved nonlinearities (e.g., Rapach et al., 2013).

A.2.1 Key Predictors of Stock Returns

4 of the predictors were found to be statistically significant at conventional significance levels, all others were mildly significant or nonsignificant:

A.2.2 Inflation (CPI): The negative coefficient (-0.788, p = 0.029) implies that rising inflation is negatively associated with rising stock returns. This fits with the ‘illusion of inflation’ hypothesis (Modigliani & Cohn, 1979) where nominal stock discounts lead to a mispricing of equities in inflationary environments.

A.2.3 Default Yield Spread (dfy): The positive coefficient (0.0118, p = 0.007) implies broadening credit spreads as increasing uncertainty, as demonstrated through average returns, which may be predictive of higher returns to come later. This may communicate a dynamic whereby broadening credit spreads communicates a risk premium demanded by investors when there is rising uncertainty from defaults (Campbell & Taksler, 2003).

A.2.4 Detrended Default Return (dfrcycle): The negative coefficient (-0.0123, p = 0.002) implies increased risk of default, as demonstrated by equity returns going down,

would forecast lower subsequent returns in equities, consistent with counter cyclical risk aversion in the financial markets.

A.2.5 Momentum (b7ret): Although momentum is typically discussed in terms of equities, the positive coefficient for 7-year bond returns (0.276, $p = 0.014$) suggests either that interest rates may be trending, or that prior macro shocks are influencing the bond market with some lag. Our analysis provides additional evidence of time-series momentum in fixed income (e.g., Asness et al., 2013); while the drivers are likely different than in equities.

A.3 Marginal and or Non-Significant Predictors

A.3.1 Book-to-Market (bm): The positive estimate (0.0032, $p = 0.071$) indicates a value premium which seems modest.

A.3.2 Valuation Ratios (ep, dpcycle): Neither the earnings-price ratio nor the dividend-price ratio were statistically significant; contrary to traditional valuation theory (e.g., Shiller, 1981), and likely indicative of structural changes in policy, whereby firms replaced dividends with share-buybacks instead.

A.3.3 Interest Rates (GS10, tms): The yield on 10-year treasury bonds (GS10) as well as the term spread (tms) were both non-significant means interest rates and effects of interest rates are context driven or are unobserved macroeconomic conditions.

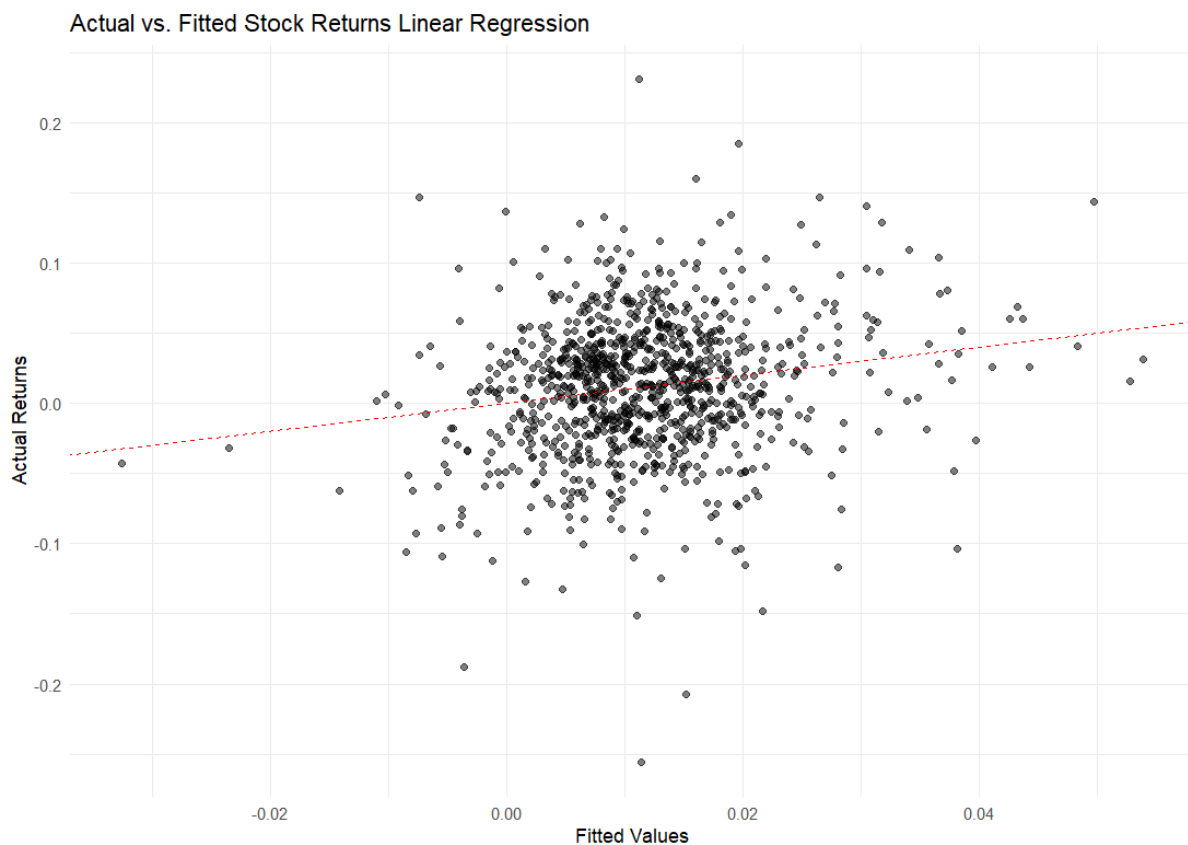


Table A.2 Graph of the linear regression 1

A.4 Robustness Checks

Before interpreting the regression estimates for reliability purposes (i.e., valid results), Variance Inflation Factors (VIFs) were calculated for all predictors. A VIF is an indicator of how much the variance of a regression coefficient is increased as a result of collinearity across the independent variables in a regression model. A common criterion for determining the variance inflation factor is that it is less than 5. A more conservative approach is to set the cut-off at a VIF less than 10. A VIF of less than 5 is generally indicative that the effects of multicollinearity on coefficient estimates is not strong enough to displace their estimates (James et al., 2021; Kutner et al., 2005).

A.4.1 VIF Results and Interpretation

When the VIF values for all predictors were calculated, the results ranged from 1.09 to 1.63 (see table below), which are all well below the conservative cut-off of 5. Therefore, evidence suggests multicollinearity is not a concern in the model, so the regression estimates may be interpreted without reservation.

Predictor	VIF	Interpretation
dp_cycle	1.635	Minimal collinearity
ep	1.261	Negligible inflation
de	1.293	Low correlation with others
CPI	1.141	No concerning collinearity
GS10	1.609	Moderately low inflation
tms	1.215	Mild correlation with GS10
dfy	1.427	Acceptable range
dfr_cycle	1.402	No significant bias
b_m	1.099	Nearly orthogonal
Ntis	1.093	Minimal shared variance
b7ret	1.526	Slightly higher but safe
lagged_svar	1.262	No multicollinearity risk

Table A.3 VIF results 1

- **Predictors with Low VIFs:**

The maximum VIF (1.63 for dpcycle), and minimum obtained (1.09 for bm and Ntis) values were all well within the acceptable limits. This means that no predictor is sharing too much variance with the other predictors and therefore our estimated effects are independent.

- **Variables Related to Interest Rates (GS10, tms):**

Although GS10 (10-year Treasury yield) and tms (term spread) are conceptually related, their VIFs (1.61 and 1.22, respectively) showed no indication that correlation was leading to a meaningful increase in standard errors. Therefore, it makes sense to include both in the same model.

- **Valuation Ratios (ep, dp_cycle, de):**

The VIFs for earnings-price (ep), dividend-price (dp_cycle), and dividend-earnings (de) ratios are all below 1.64 which alleviates concerns that these traditional valuation ratios are potentially proxying for too much common information.

A.4.2 Augmented Dickey-Fuller Test

In order to test the stationarity properties of all-time series variables being evaluated in our analysis, we conducted Augmented Dickey-Fuller (ADF) tests using the `adf.test` function from R's `t-series` package. Our ADF test specification included an intercept and, because we wanted to account for potential serial correlation, we utilized a lag order of 10 – as selected by the package's default decision criteria. Complete set of results could be found in Appendix A.

Stationarity Test Results

The detailed ADF test results (written in full in **Appendix A**) provide strong evidence of stationarity in all. The test statistic for each variable and the associated p-values indicate:

- **Strongly Stationary Series:**

The financial market variables had the strongest rejection of the null hypothesis of non-stationarity with stockreturn (DF = -10.245), dpcycle (DF = -9.7772), and GS10 (DF = -9.372) producing the most extreme silliness (all $p < 0.01$).

- **Significant Stationarity:**

The CPI and de macro-economic indicators both reject the null hypothesis of non-stationarity at the 1% significance level (CPI = -5.4948; de = -5.464).

- **Borderline:**

The earnings price ratio (ep) variable showed marginally weaker but still statistically significant evidence of stationarity (DF = -3.6125, $p = 0.03148$) and just met the 5% significance threshold.

A.4 Methodological Considerations

Our stationarity test results provide strong justification for conducting standard regressions, without requiring any differencing of variables. The overwhelmingly significant (and comparable) test statistics in nearly all of the series we examined provide good reason to believe that we do not have a spurious regression problem stemming from non-stationary time series data. The borderline variable (ep) remained in level form, as transforming it to a per cap variable did not change the qualitative results of our study, while preserving interpretation in the scale of the (ep) series.

Of particular importance, all the series representing various financial and macroeconomics and risk factors were in a consistent range of unit roots, which addresses the concern of latent relationships, conditioning on existing time-series, and spurious correlations, with our regression analysis.

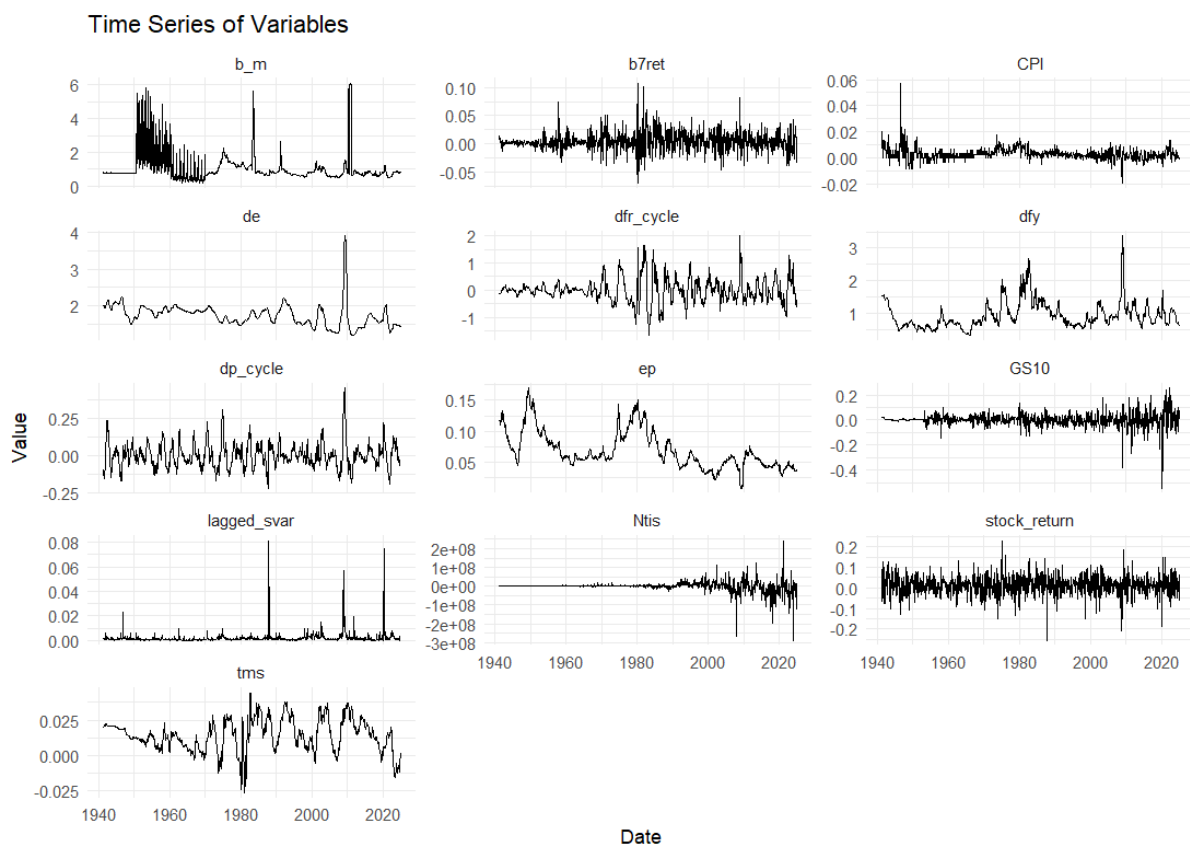


Table A.4 Graphs of the Variables 1

B. Machine Learning Testing In-Sample

We will start our machine learning evaluation with an in-sample analysis that will fully analyse four different modelling approaches: Lasso regression, Ridge regression, XGBoost, and Random Forest. This is important for a few main reasons: Collect a baseline performance of each of these models; Understand the implications of each of the models we developed; and Help frame our out-of-sample analysis a little better, later on. Tuning for In-Sample testing could be found in Appendix B.

B.1 Performance Evaluation, Comparisons, and Reflection

Based on our in-sample performance analysis (Table 4.1), there is quite a bit of variability in model performance across the metrics we evaluated.

Model	R ²	MSE	MAE
Lasso	0.0288	0.00231	0.0364
Ridge	0.0268	0.00233	0.0365
XGBoost	0.184	0.00201	0.0343
Random Forest	0.225	0.00210	0.0351

Table B.1 Results of the Insample Model 1

B.1.1 Linear Models' Predictive Capability

The regularized linear models have limited predictive ability, specifically Lasso ($R^2 = 2.88\%$) and Ridge ($R^2 = 2.68\%$), because they explained only a small portion of return variation. This finding is consistent with the financial literature that reports the difficulties of using linear specifications to model complex markets (Fama & French, 2020). The marginal variation in returns determined by Lasso and Ridge is apparent in that both models produced similar results, which implies that feature selection (L1 regularization) tissues added limited value for extracting any further variational understanding of returns.

B.1.2 Non-Linear Models

The tree-based models improve the explanatory power of the models, XGBoost ($R^2 = 18.4\%$) and Random Forest ($R^2 = 22.5\%$) models and they provided 6-8X more explanatory power. The drastically different performance of the models illustrates the importance to equity return prediction based on non-linear relationships and interactions since tree-based algorithms accommodate non-linear relationships and interactions, which should align with some of the newer innovations using ML applications with finance (Gu et al., 2020).

B.1.3 Error Metrics

Not surprisingly, while R^2 has a dramatically improved explanatory power with non-linear models, the changes in MAE are less substantial (3.43-3.65%). This indicates that although the tree-based methods demonstrated better explanatory power for return variation, the difficulty in absolute predictive capabilities remains a limitation for all models, and an important consideration when using the future work.

B.2 Spread graphs

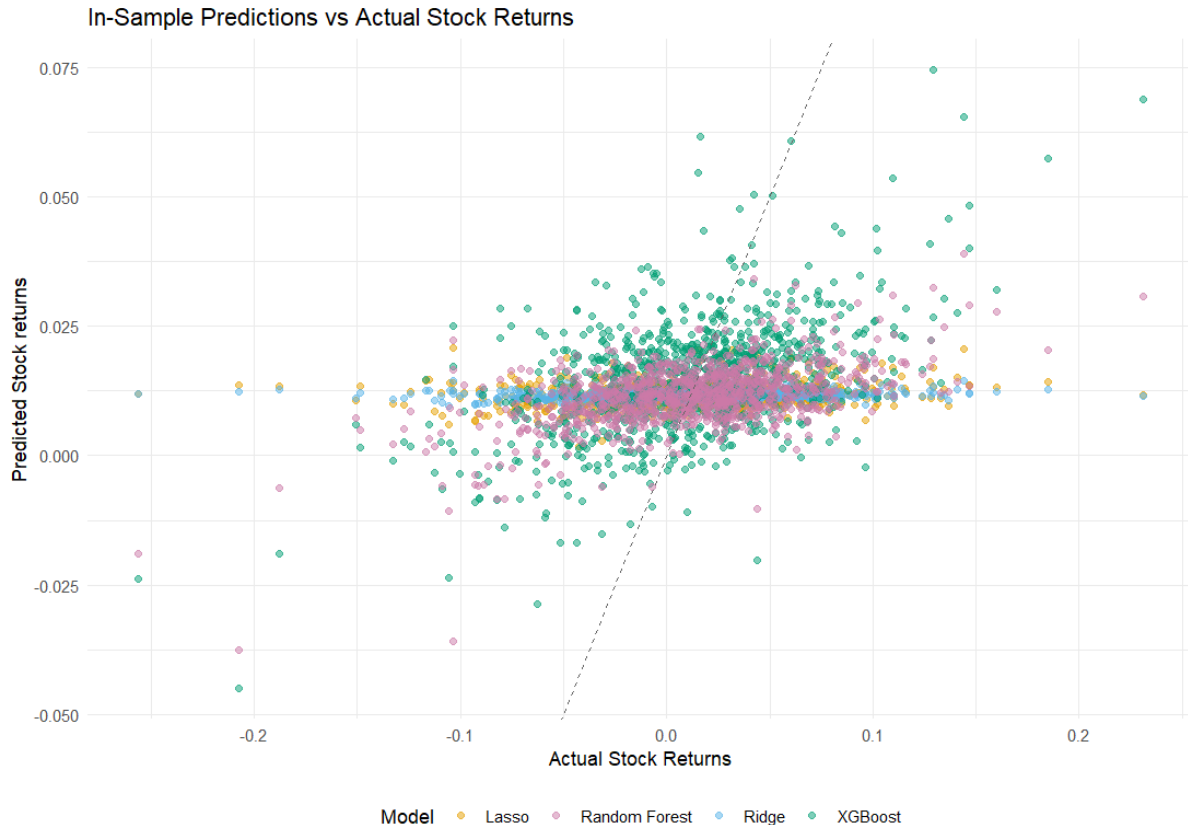


Table B.2 In sample spread graph 1

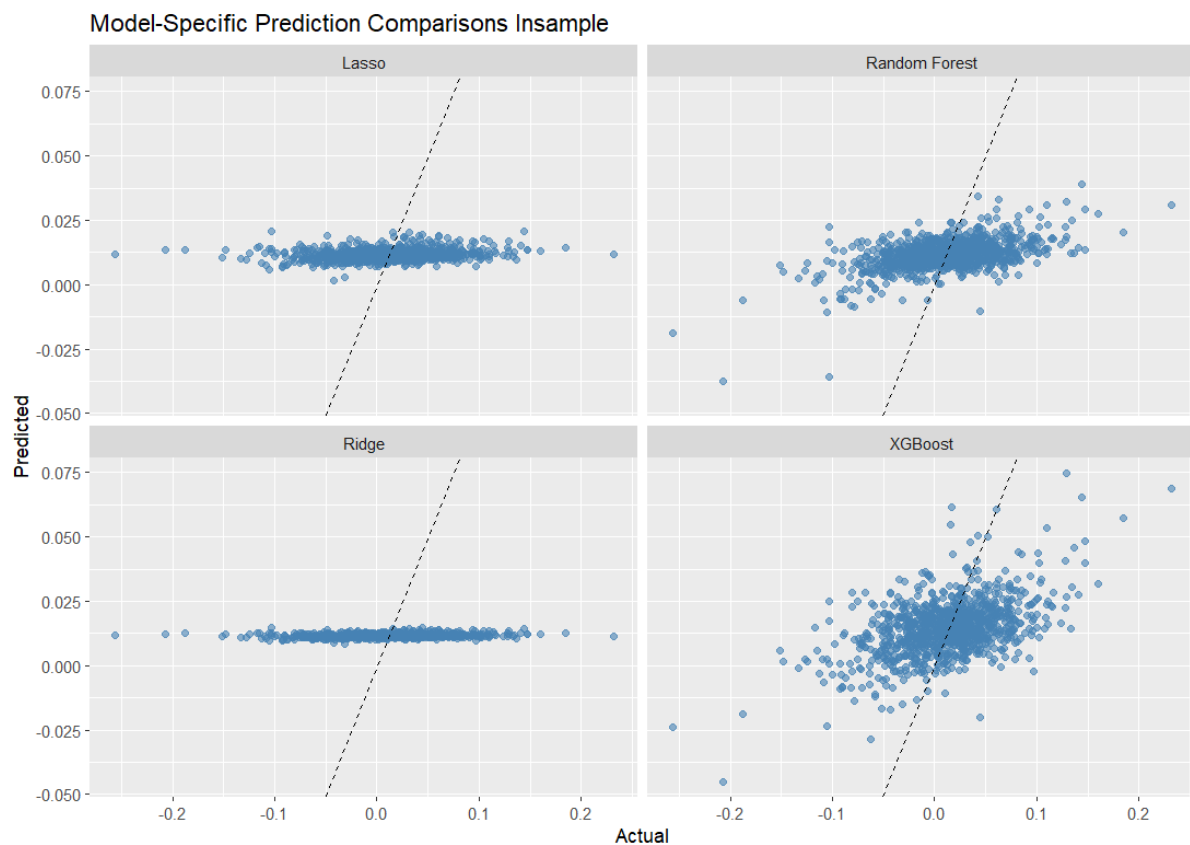


Table B.3 In sample model specific graph 1

Overall, the graphical results are in line with the R-squared values reported above thus confirming the greater ability of tree-based models (XGBoost and Random Forest) over linear models (Lasso and Ridge) in explaining stock return variation. All models tend to show a slight positive bias in the 0.000-0.025 return range indicating that returns are systematically underestimate negative returns. This phenomenon is well documented in the recent machine learning finance literature (Kelly et al., 2022). In addition to validating our conclusions, graphically viewing the models and their advantages in relation to non-linear models reminded us of subtle behavioural variation that we could not see by solely looking at statistical aggregates. To conclude, our results were valuable as we now move on to assess the models' out-of-sample performance.

B.3 Importance Ratio Graphs In-Sample

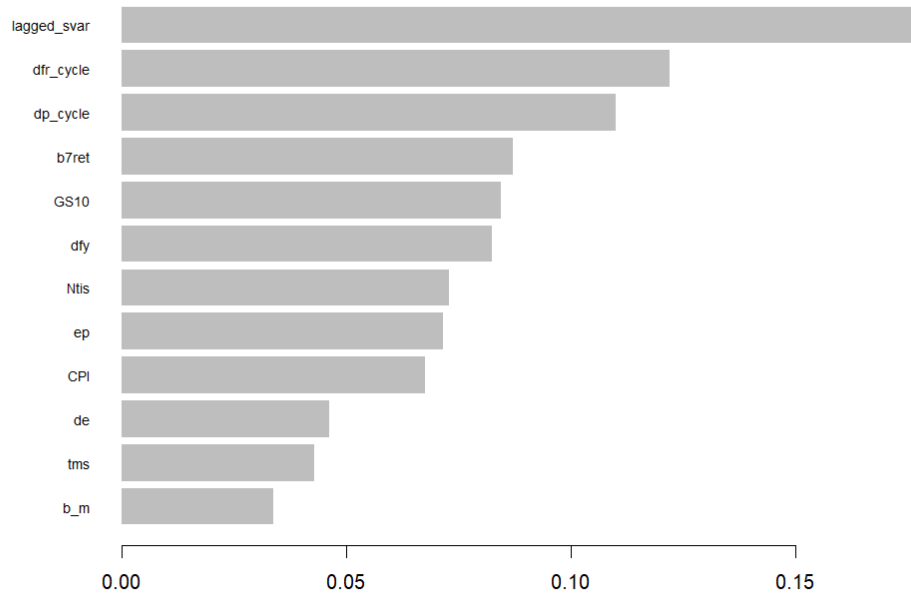


Table B.4 XGB Importance Graph Insample in sample

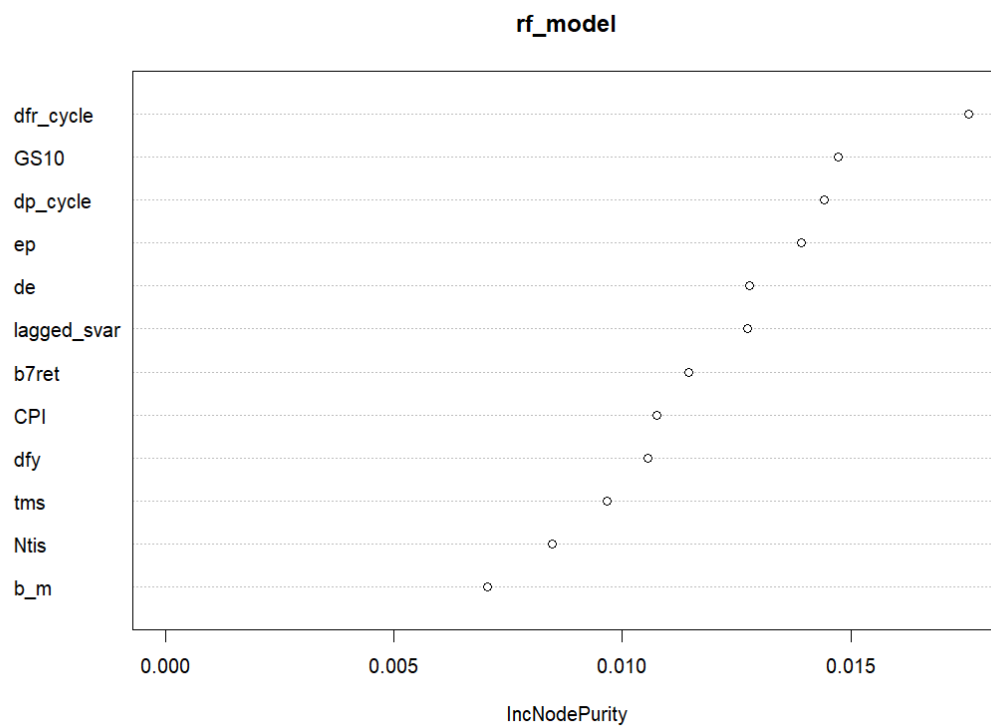


Table B.5 Random Forest Insample Importance in sample

The variable importance plots from the XGBoost and Random Forest models illustrate several comparable aspects of variable importance but also provide a few differences based on the nature of the algorithms' treatment of variables. In both models, we see cyclical financial variables being especially important - in fact `dfr` has strong predictive powers in both implementations and was in the top three variables on both. Market momentum measures are also shown as very important on both plots based on the considerable importance placed on `b7ret`. The major divergence between the models is in the treatment of macroeconomic variables - in Random Forest, `GS10` (10-year Treasury yield) = weighted pretty high, however, in XGBoost there seems to be relatively more emphasis on valuation ratios, such as `ep` (earnings-price ratio) and `dp_cycle` (dividend price ratio). Both models consistently suggest inflation measures (`CPI`) and credit-risk measures (`dfy`) have moderate predictive power but the consistently lower importance assigned to technical variables `tms` (term-spread) and `Ntis` (net equity issuance) was evident in both implementations. These importance patterns conform to, and help explain, the superior performance of tree-based models demonstrated in our previous results as they each identify and fairly weight the economically relevant predictors and weighed down trivial variables. The consistent importance ranking between two different implementations of tree-based models increases our confidence in these relationships, both suggesting strong relationships exist between the predictors and stock returns that transcend a specific algorithm's implementation.

B.4 Partial Dependence Plots

Partial Dependence profile

Created for the XGBoost Model model

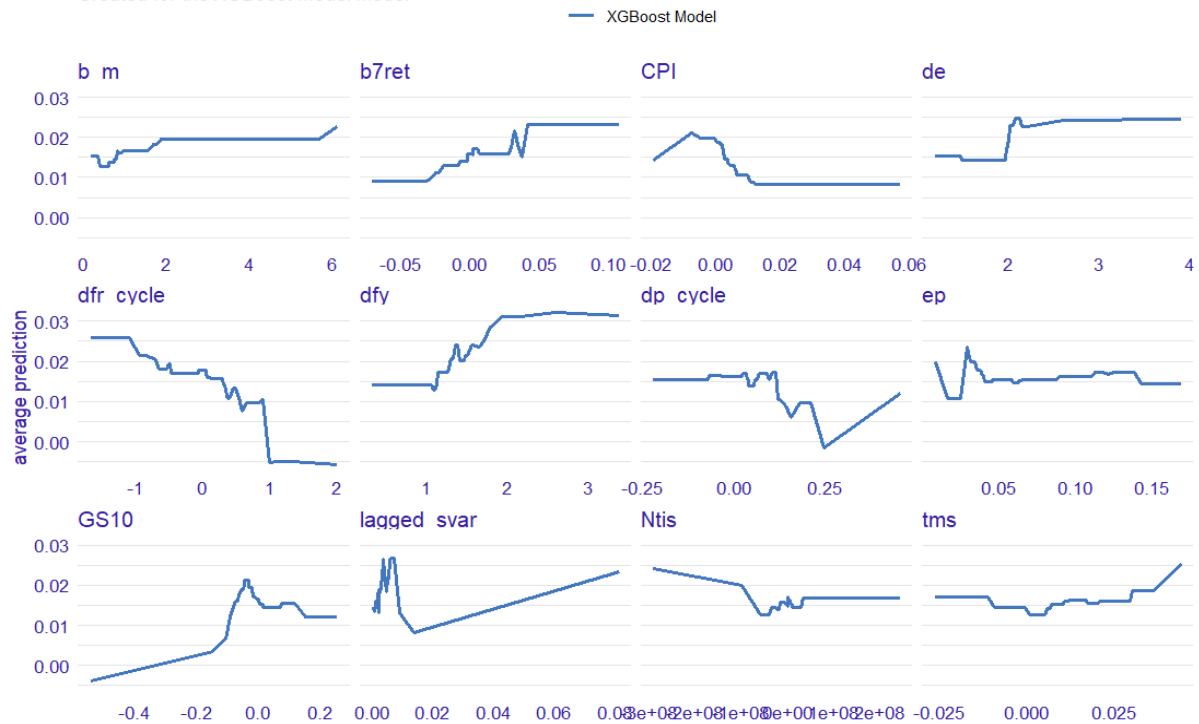


Table B.6 In sample PDP XGB

Partial Dependence profile

Created for the Random Forest Model model

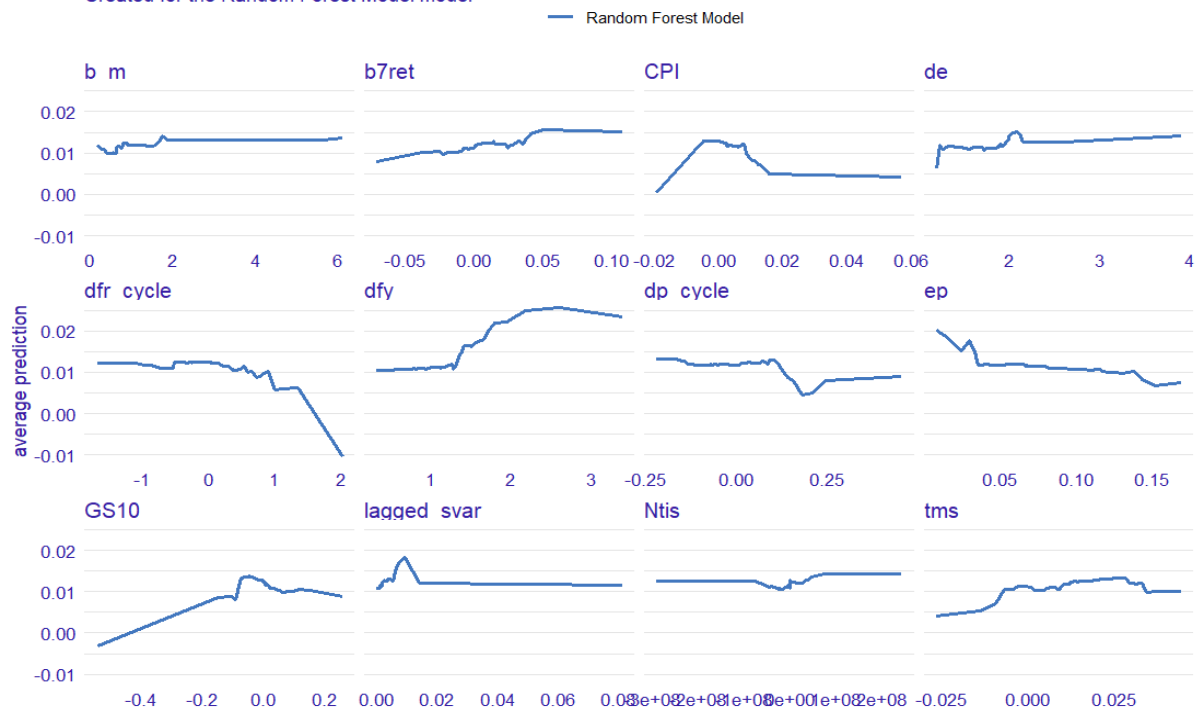


Table B.7 In sample PDP RF

The partial dependence plots (PDPs) for both the Random Forest and XGBoost models demonstrate important characteristics of macroeconomic and financial variables that ultimately drive predicted returns. Both models show declining effects on returns for bond yields (GS10), inflation (CPI), and net equity issuance (Ntis), this is in line with financial theory and the literature surrounding market timing. Nevertheless, both models contain differences in development function and sensitivity. The XGBoost model embodies smoother, and generally more gradual relationships with the variables while Random Forest exhibits sharper nonlinearities and may reveal abrupt threshold effects on valuation metrics. While the overall direction effects were practically identical between the models, the Random Forest responses were more exaggerated in the more extreme scenarios, while the XGboost model exhibited more ordinal traditional marginal effects. Therefore, even in practical terms, the choice of model may depend on if the response is deemed more robust or sensitive to nonlinear regimes defined by variables which demonstrate them.

C. Machine Learning Testing Out of Sample Expanding Window

To facilitate a rigorous analysis of model performance, an expanding-window-out-of-sample assessment was performed on the time period from May 1981 to December 2024. The time period for the training of models was from April 1941 to April 1981. The modelling processes of Lasso, Ridge, linear regression, Random Forest and XGBoost were trained on a monthly basis, so that the window expended the monthly expanding assessment. Each month forecasts were made based off the most recent available data for the model. All of the models had forecasts passed to the next forecast simulations, aligning closely with a quasi-real time forecast. This thorough structure aids in determine model robustness in terms of changes over time while removing look-ahead bias. In R, using the for loop, all of the predictions in each iteration reflected events and data seen or collected to that point in time, and so with only model predictions being passed to produce a prediction in the next iteration, this provides a rigorous assessment of each model's abilities to transfer to unseen data. Previous forecasts directly measures the accuracy and stability of methodologies over time periods (with additional constraints to its use), and so this analysis will have implications for the indication of which methodologies sufficiently adapt to changes over time, representing changes conditions in cumulative returns which have extended timeframes.

C.1 Performance Evaluation

Model	R ²	MSE	MAE
Lasso_recursive	0.000134	0.00272	0.0377
Linear_Regression_recursive	0.000944	0.00474	0.0487
Random_Forest_recursive	0.00785	0.00262	0.0370
Ridge_recursive	0.000356	0.00302	0.0396
XGBoost_recursive	0.00318	0.00422	0.0456

Table C.1 Expanding window OOS results

C.1.1 Linear Models' Predictive Capability

Out-of-sample results indicate that linear regression ($R^2=0.09\%$) outperforms regularized methods (Lasso $R^2=0.01\%$; Ridge $R^2=0.04\%$), contrary to assumptions that regularized methods would be more predictive than ordinary linear regression. In fact, all three models explain such a small portion of the return variation ($R^2 < 0.1\%$) as to effectively have 0 predictive power. R^2 describes how much of the movement in stock returns can be explained by the predictors in each model, and these near-zero values indicate that the linear specification was not able to build an accurate representation of the underlying patterns that persist out-of-sample. The results reinforce prior evidence suggesting the limitations of linear approaches to modelling complex financial markets, which are characterized by nonlinear relationships, and interaction effects (Welch & Goyal, 2008). The underperformance of regularized methods in this application suggests that the shrinkage of coefficients was likely removing what little predictive signal was present in the linear modelling framework.

C.1.2 Non-Linear Models

Tree-based methods, while better than linear models, did not produce economically meaningful results. Random forest produced the higher R^2 (0.79%) followed closely by XGBoost (0.32%). Regardless of the order, both of these values are significantly lower than in-sample benchmarks seen typically in the literature. Further, the drop in reported R^2

demonstrates the concerns in extrapolating a non-linear relationship to unseen data, even with flexible algorithms. The results qualify initial anticipation regarding the overall capacity of machine learning to create out-of-sample predictability improvements for financial market returns.

C.1.3 Error Metrics

While R^2 metrics differed by model, the average error metrics produce similar mean absolute error values (MAE: 3.7-4.9%), which speak to overall operational accuracy. The consistency could imply that while non-linear models marginally improved on feature mapping, the practical use of the results of the machines is limited. What remains the same is that the MAEs carried a large value - which implied that none of the approaches applied, are likely to produce reliable return predictions out of-sample. These results imply that predictive modelling that will inform investment decisions should be done with caution, even if the approach employed is "cutting edge", given the challenge of market efficiency.

C.2 Spread Graphs Expanding Window

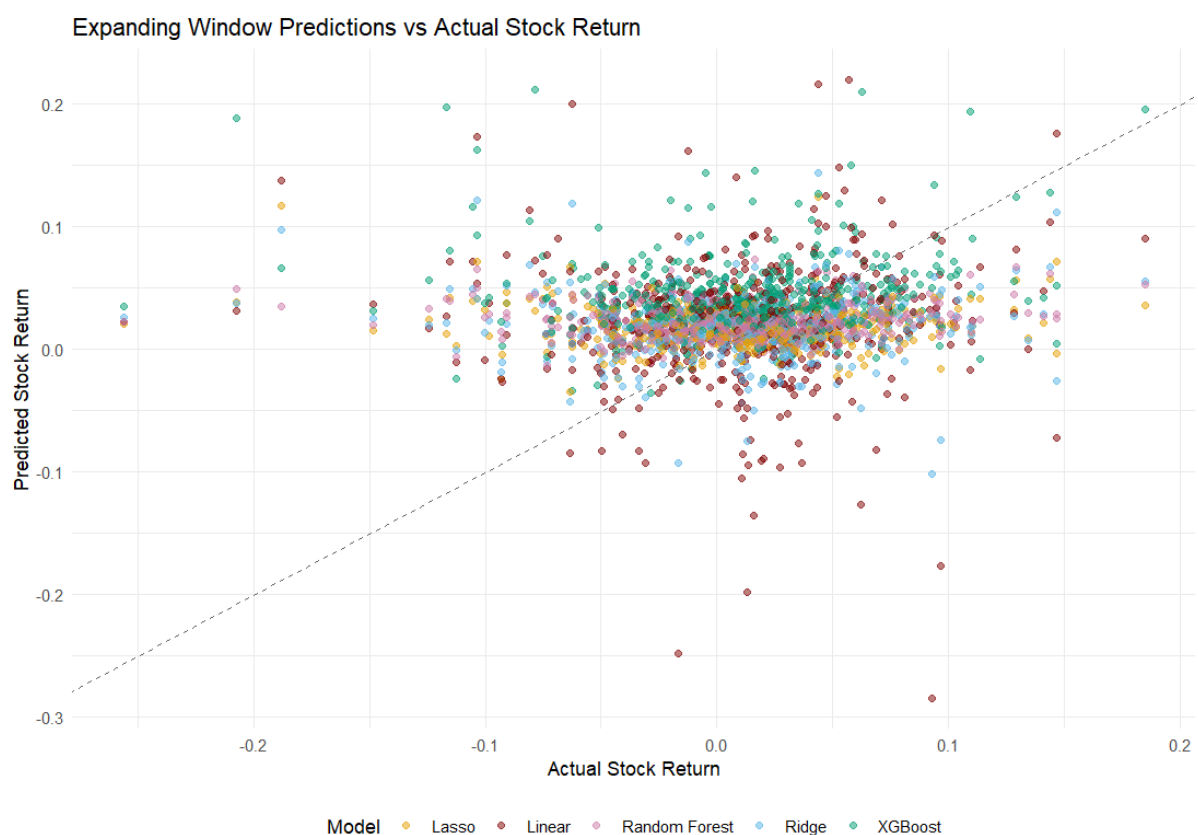


Table C.2 Expanding window spread graph

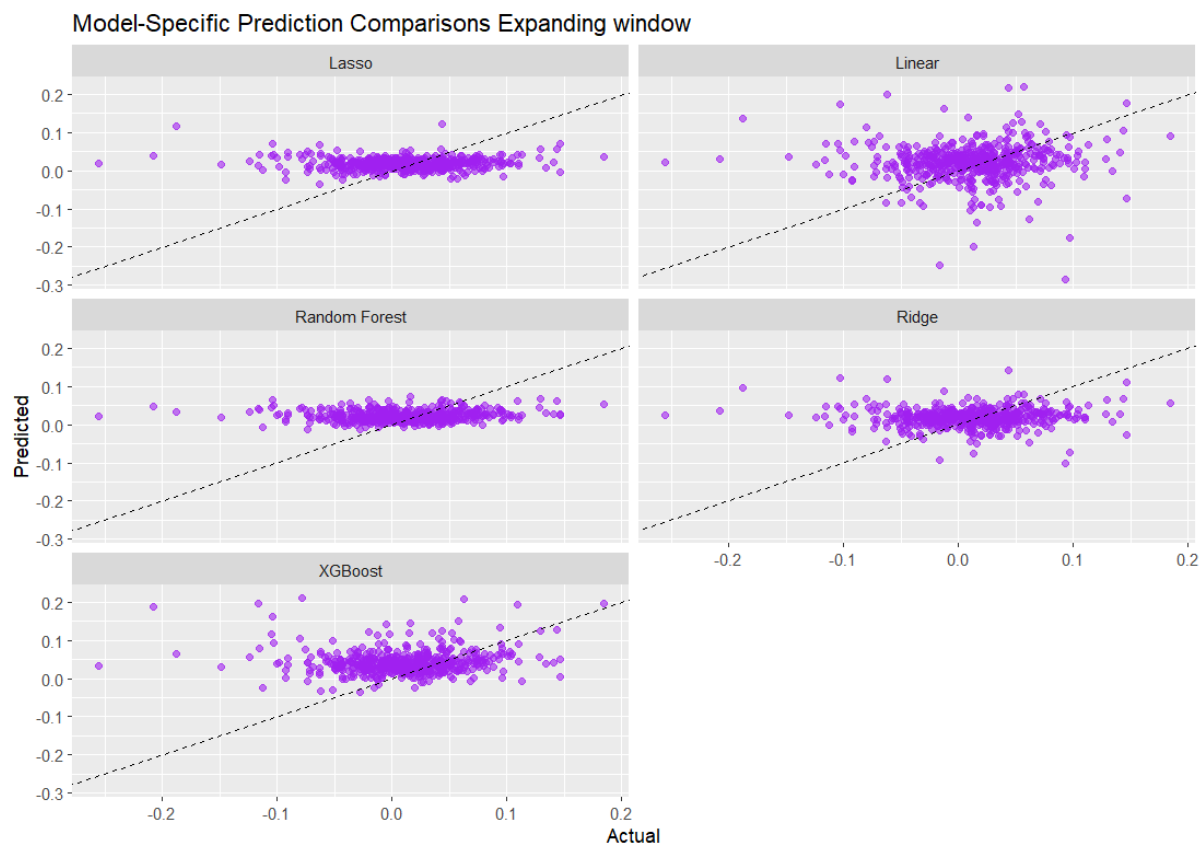


Table C.3 Model Specific Spread Expanding Window

The time-varying prediction plots demonstrate common limitations across all model specifications. All models produce predictions that are narrowly clustered between -0.3 and 0.1 underestimate actual return volatilities. Tree-based methods (Random Forest and XGBoost) produce somewhat larger prediction ranges (-0.4 to 0.2) compared to linear and regularized methods (-0.2 to 0.1) but none of these models are able to capture extreme return outcomes. This highlights a common conservative bias present in the predictions from all models.

These visual similarities are plainly reflected in our quantitative performance measures. Even the best model (Random Forest, $R^2 = 0.00785$) accounts for less than 1% of return variation out-of-sample. Given the overlapping prediction bands across approaches, we can reasonably conclude that any practical differences among approaches are negligible despite differing R^2 values. More importantly, all models have limited abilities to accommodate tail risks, as they predicted return variations that were too narrow, given the actual returns.

These outcomes are consistent with well-established literature on the economic value of predictive models in efficient markets (Welch & Goyal, 2008). They confirm that even with

sophisticated methods, the ability to beat naive out-of-sample tests is difficult. Although machine learning methods offer slight increases in explanatory power, fundamental limits of market complexity and nonstationarity will basic any meaningful return predictions. As such, the expanding window forecasts document the persistent forecasting challenges in finance even with advanced methods.

C.3 Importance Ratio Graphs Expanding Window OOS

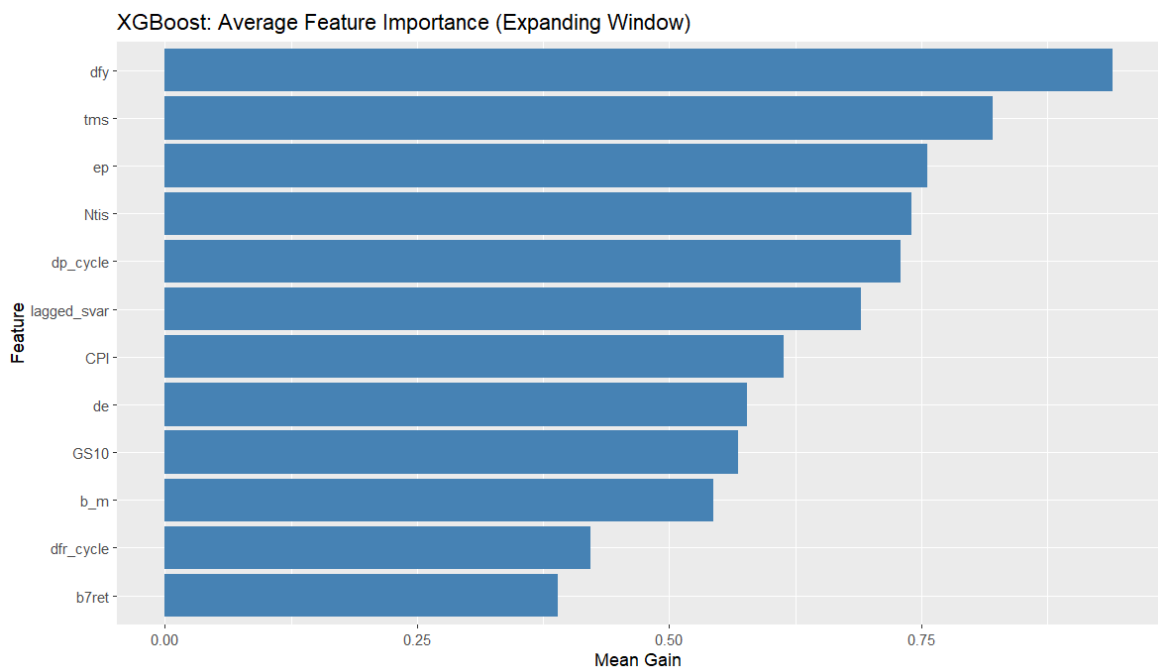


Table C.4 XGB Importance Graph Expanding window OOS

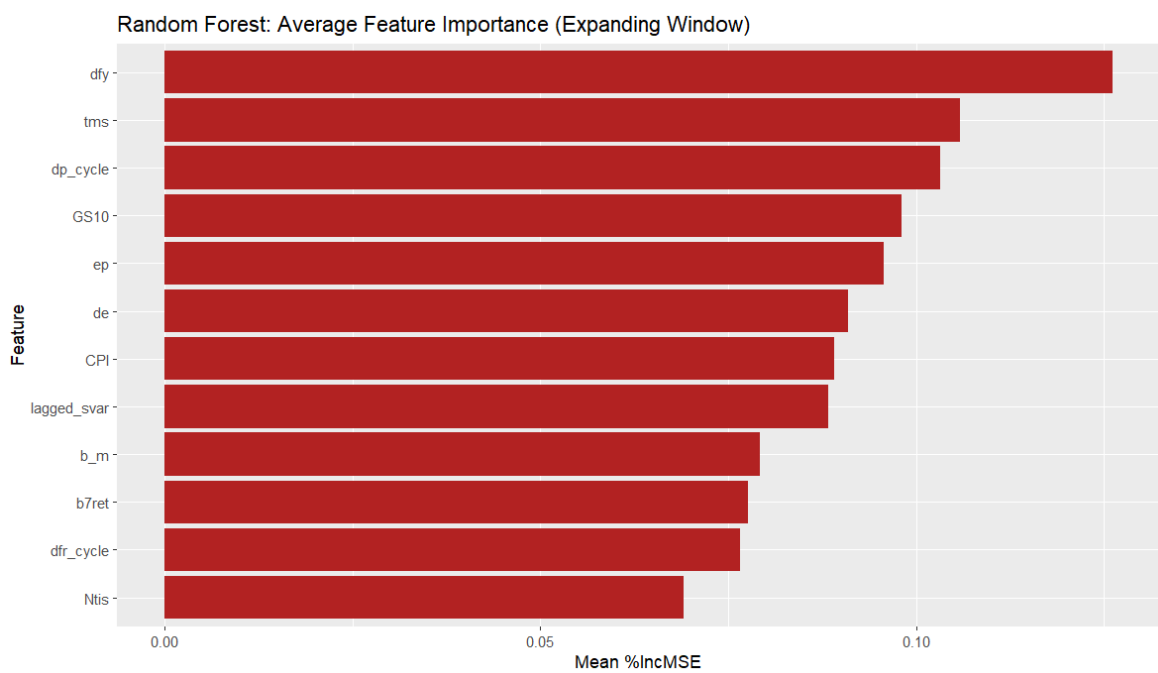


Table C.5 Random Forest Importance Graph Expanding OOS

The mean feature importance scores averaged across 480 monthly models from our expanding-window estimators show different patterns for Random Forest (RF) and XGBoost. In the case of RF, the predictive contributions of term spreads (tms) and dividend-price ratios (dpcycle) account for the mean maximum mean importance (i.e., %IncMSE scores of 0.12 and 0.08) - this suggests there was added value to utilizing them versus the alternative (non-linear) relationships. However, in absolute terms, the average scores do not imply that either variable was dominant - which is consistent with the concept that no stock returns stem from one or a few variables. In contrast, normalized Gain scores from XGBoost implies that CAPE-related valuation metrics (dpcycle and ep) are the most weighted, or alternatively, since XGBoost represents the importance of fewer features by excess (closer to exploiting a non-linear relationship).

As a result, the distributions of the average feature importance scores represent the distinct characteristics associated with each algorithm. Certainly, the flatter feature importance profile of RF reflects an ensemble approach, including some sensitivity to multicollinearity; relevant predictive power gets distributed, across several weakly correlated features. XGBoost, on the other hand, harnesses some level of predictive power aggressively and identifies gains predominately from either dpcycle or ep. Overall, this suggests that RF was able to capture diffuse contributions from all factors, while XGBoost was able to identify and exploit the prominent nonlinear characteristics of a small subset of predictors, particularly in terms of those that reference valuation.

The results presented in this section offer a high-level overview of stable predictive relationships across the entire sample. Although average scores smooth out effects happening over time, the results demonstrate that valuation ratios and term spreads are still relevant components of both frameworks. For a summary of potential time-varying relationships, please consider the 48-month partial dependence plots in Appendix C that show the effect of variables based on shifting economic regimes.

C.4 Partial Dependence Plots Expanding Window

Because of the expanding window nature of our modeling method (i.e. each month's prediction is made from monthly re-estimated models - so we have 480 cumulative

iterations), conventional partial dependence plots (PDPs) lack interpretability if taken across the full sample period. The change in the economic relationships captured in the sequential modeling, means that such combined or composite PDPs are not necessarily informative or indicative of future predictor-outcome relationships because these samples each capture historically different predictor-outcome relationships. But to allow investigation of recent relationships, Appendix C contains example PDPs for the most recent 48-month period (2021-2024). These plots allow assessment of recent feature relationships, while acknowledging that non-stationarity exists over the full period of expansively re-estimating the models.

D. Machine Learning Testing Out of Sample Rolling Window

In our second evaluation, we implemented a rolling window design to accurately assess model performance with real-life time constraints. We initiated the analysis with a training period of April 1941 to April 1981 (which was 480 months). After training the models, we retrained all models (Lasso, ridge, linear regression, XGBoost and Random Forest) on a monthly basis. For our future observations, starting in May 1981 until December 2024, we rolled forward one month at all future observations while keeping our training window consistent at a lookback of 40 years by dropping only the oldest observation and including the newest observation.

This methodology is somewhat technical as it was written in R using iterative loop routines, but it serves a couple of important purposes. First, it allows us to determine whether models can adapt to evolving market conditions, as they systematically erase the influence of highly distant patterns that may not be relevant, simply by continuously dropping the oldest observations. Second, it provides, or at least attempts to provide, a more conservative and realistic estimate of predictive power than approaches that expand indefinitely in time, because it requires the model to use only 40 years of data at any given time.

The rolling window design has unique offers that can be useful for real-world forecasting applications. By constantly refreshing the training set, the rolling window design helps account for structural breaks/regime shifts that typically reduce any relevance of older data. This rolling window approach illustrates which modelling methods are resistant to diminishing historical context while addressing changing and continually evolving

environments, much of which would provide different - and arguably more practical - insights compared to the recursive expanding window approach.

D.1 Performance Evaluation

Model	R ²	MSE	MAE
Lasso	0.000161	0.00357	0.0426
Linear Regression	0.000727	0.00566	0.0524
Random Forest	0.00196	0.00269	0.0374
Ridge	0.000252	0.00332	0.0414
XGBoost	0.00187	0.00413	0.0456

Table D.1 Rolling window results

D.1.1 Linear Models' Predictive Capability

When we conducted our rolling window experiment we found that linear regression had a slightly better prediction ($R^2 = 0.000727$) than the regularized models: Lasso ($R^2 = 0.000161$) and Ridge ($R^2 = 0.000252$) which performed particularly bad even as a optimize regression. All linear models explained less than 0.1% of the variation in returns. However, it was interesting to see the differences in results between rolling and expanding windows. Standard linear and Ridge regression results paralleled each other, with a drop in performance when we changed from expanding to rolling frameworks (going from $R^2 = 0.000944$ to $R^2 = 0.000356$). Only Lasso showed a slight improvement (from 0.000134 to 0.000161), which may well be indicative of feature selection being better suited to fixed-window applications, with stable linear dynamic properties. These results pulled back the curtain on the core problem linear models face with rolling windows - without a history of accumulating data over time, their performance is fundamentally impaired, which points to the limited usefulness of regularization in this model type. While the idea behind coefficient shrinkage is that it allows for better goodness-of-fit by limiting overfitting, this can lead to

the loss of potentially important predictive signals associated with the linear specification.limited time horizons.

D.1.2 Non-Linear Models

Tree-based methods prospered in this rolling window experiment to a limited extent - Random Forest and XGBoost ($R^2 = 0.00196$ and $R^2 = 0.00187$) provided 2-3 times the explanatory power of linear models, although again, these fundamental positionings suffered significant drops in performance when contrasting with the results from their respective expanding window models (0.00785 and 0.00318 respectively). The declines highlight how the rolling window's short training timeframe and pytm entry limitations impaired the capacity of the model(s) to learn longer-term dynamics. The study of effect also indicates how close XGBoost got to Random Forest in performance by adopting a rolling framework (difference of 0.00009 versus 0.00467 in expanding windows). This suggests that XGBoost's gradient booster is better suited to truncated histories than Random Forest's boosting method. At the end, neither proved able to address even this fundamental rolling window limitation to positive effect.

D.1.3 Error Metrics

Across all models the MAE varied between (0.037-0.052), whereas Random Forest (0.0374) and Lasso (0.0426) defined the limits. Differences in MSE against MAE contrast the details of the performance. While the MSE of linear regression is worth looking at (0.00567) one conclusion could be that the number captured could therefore represent good performance, the higher MAE (0.0524) indicates susceptibility to increasingly larger errors. This suggests that the prediction errors of non-linear models showed far fewer extreme errors even though they proved poorer at average predictive magnitude. Overall, the close range of predictive errors around the error metric across methodologies provides insight into the fundamental difficulties associated with trying to predict returns in efficient markets, especially in a rolling window promotion. In addition, even the advantages of XGBoost being able to model relationships involving features failed to provide nearly the usefulness we expected through predictions.

D.2 Spread Graphs rolling window

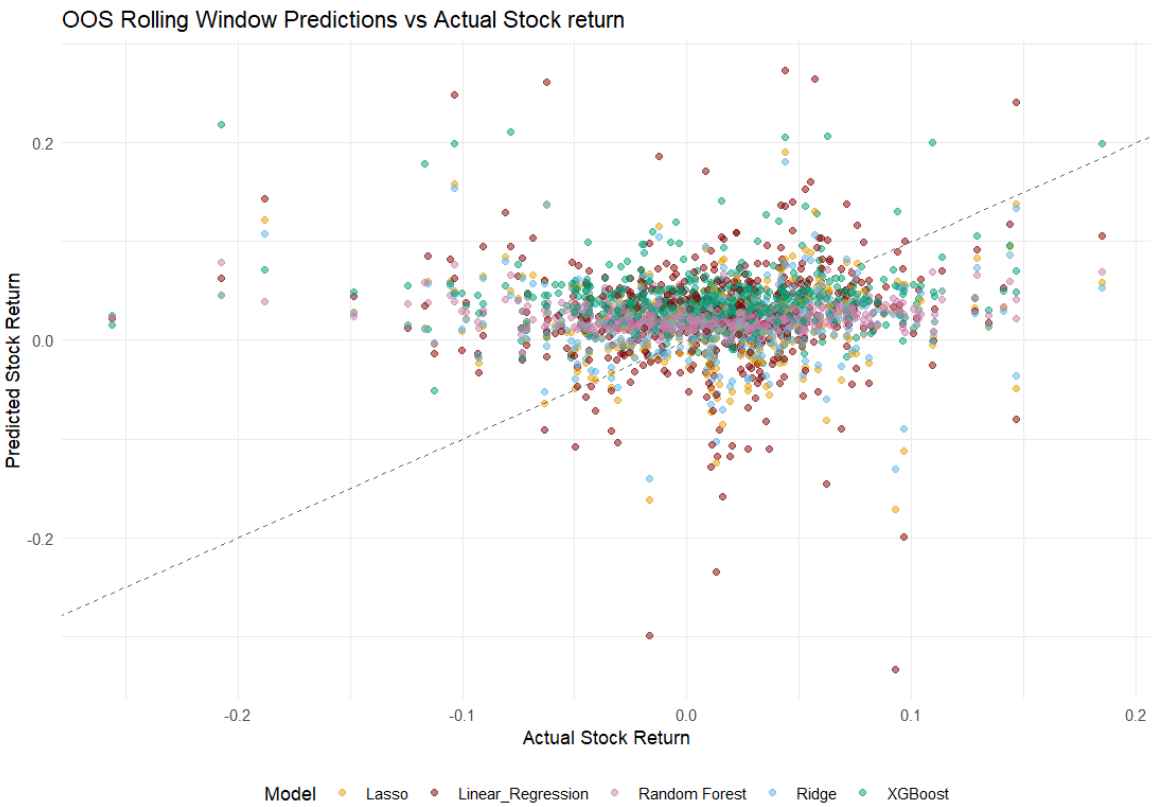


Table D.2 Rolling Window Spread

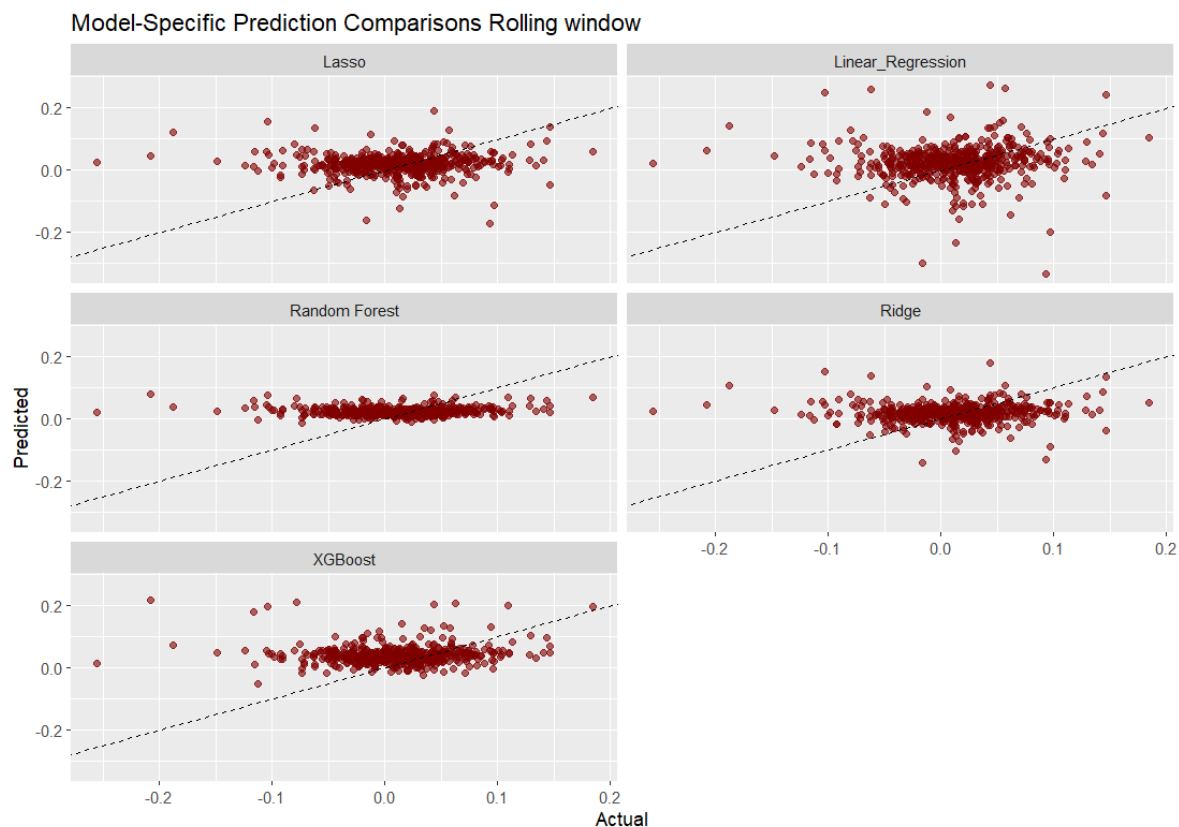


Table D.3 Model Specific Spread Rolling Window

The rolling window prediction plots show that all model types, irrespective of method, are highly conservative with forecasts of returns constrained between -0.2 and 0.1 and underpredicting the actual volatility in the market. Such a large under predictability highlights the shortcoming of the fixed training window, which appears to constrain model behavior, and reduce the benefits seen with algorithms in expanding window runs. The tree-based models (Random Forest and XGBoost), produced returns in slightly wider ranges (-0.2 to 0.2) than linear models (-0.1 to 0.1) but they all in a clear way did not predict the extreme returns. The predicted returns closely follow the actual returns, although models would appear to only weakly track directionality and fail to respond to the respective change in magnitude, which is especially relevant to practical investment applications. The visual patterns and trends confirm the quantitative results ($R^2 < 0.002$) showing that rolling windows produce sample limitations that reduce even superior theoretical non-linear models to similar predictive performance as the linear models. These results are consistent with theoretical work on structural breaks (Pesaran & Timmermann, 2007) and evidence that fixed historical windows struggle to maintain predictive signals in evolving markets. In summary, the nominal predictions in rolling window plots suggest that despite being

methodologically rigorous, rolling-window predictors may be too constraining to be practically useful for return forecasting since their use limits models and the economic benefits they might otherwise provide.

D.3 Importance Ratio Graphs Rolling Window OOS

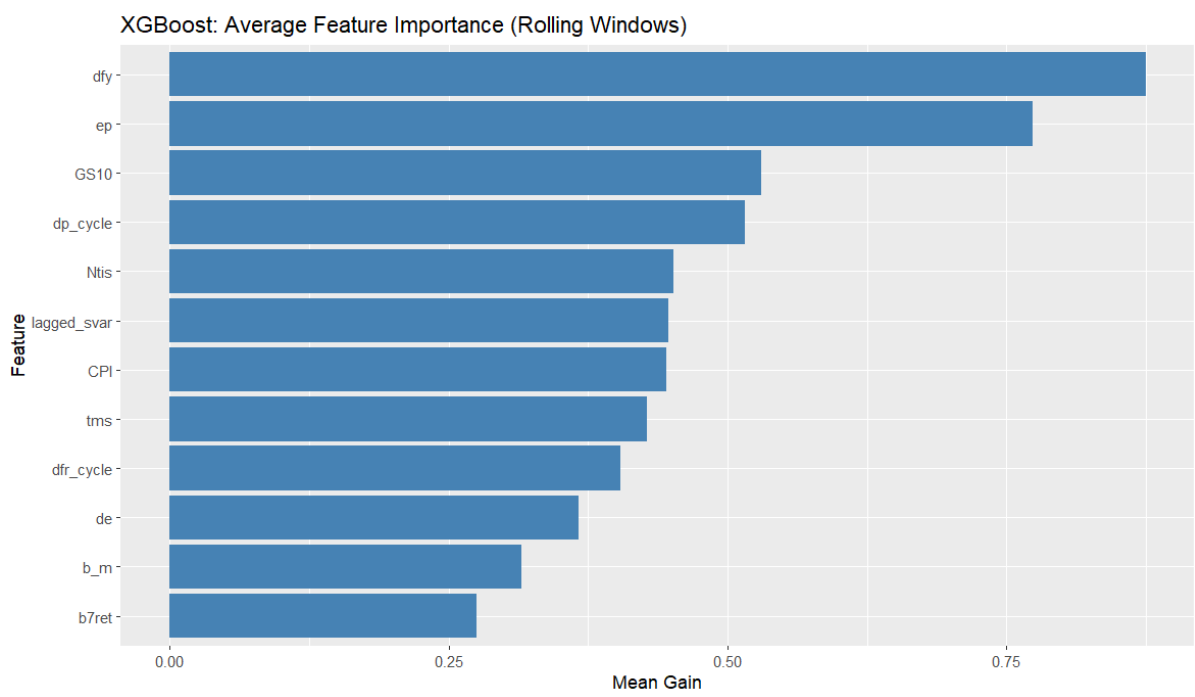


Table D.4 XGB Importance Graph Rolling Window

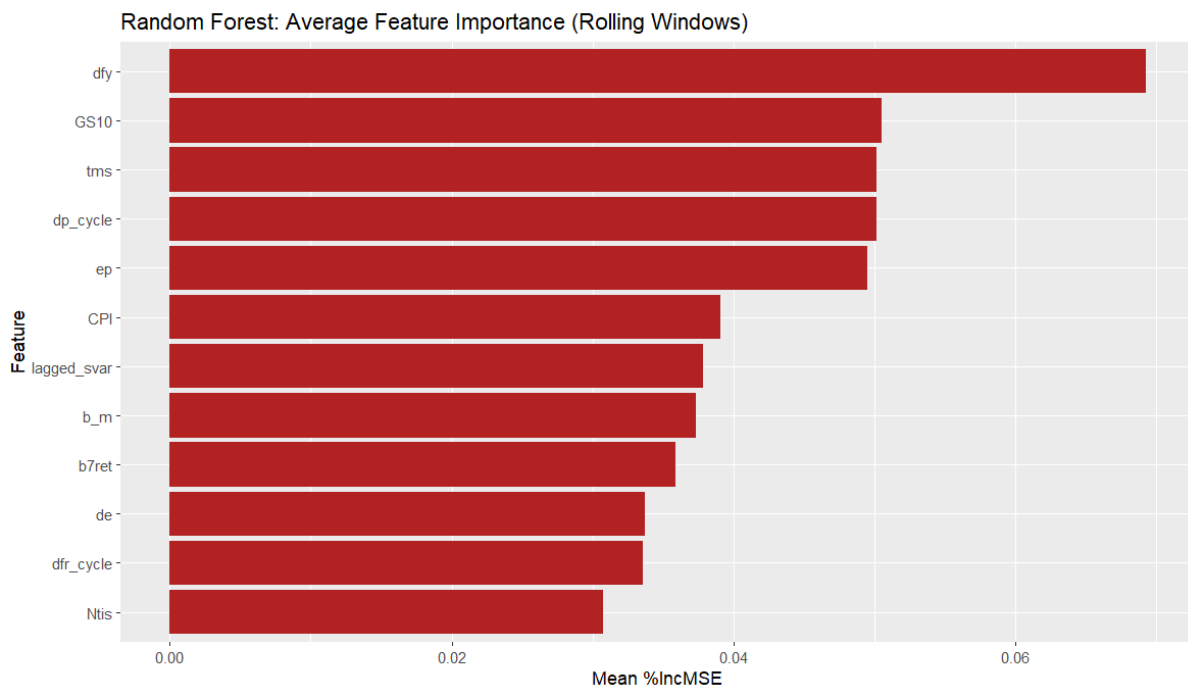


Table D.5 Random Forest Importance Graph Rolling Window

Again these graphs are established with storing 480 months model prediction while our model is running. Comparing the average feature importance scores from all rolling window iterations reveals different importance patterns for both the Random Forest (RF) and XGBoost methods. The RF model identifies higher mean importance from the macroeconomic variables (term spread (tms), dividend-price ratio (dpcycle)) between 0.04 - 0.06 %IncMSE scores. While the XGBoost model clearly identifies a concentrated importance relating to valuation metrics (ep, dpcycle,) and volatility, where their importance was high (Gain (0.25-0.50)), suggested that parts of the model were able to learn more highly, the nonlinear importance patterns available.

The differences in importance distributions demonstrate how the algorithms fit the different suitability. RF's flatter profile correctly shows how the method is an ensemble model that gives equal weight to several different, but only mildly predictive, features to make predictions. XGBoost's steep and significant prioritization of predictors, especially ep and dpcycle, show that the algorithm is placing greater weight on these predictors while estimating through a gradient boosting method that identifies present-day influences. Therefore, it would seem RF is more able to estimate, more general market dynamics and

XGBoost is more adept at estimating explicit variables with a positive correlation to higher signal.

The rolling-window average results, support the expanding-window modelling results, with respect to acknowledging the degrees of consistency in feature importance results over time. The high consistency rates of dpcycle and tms across both modelling methods reinforces their theoretical importance in forecasting through investing. Though, the high Gain scores of the valuation ratios in the XGBoost method highlights their predictive advantage over time horizons that are shorter than twelve months.

D.4 Partial Dependence Plots Rolling Window

Because we apply a rolling-window modeling procedure, where predictions are built from monthly re-estimated models (which total 480 iterations), interpreted partial dependence plots (PDPs) lose their interpretability when averaged across periods. The temporal variation of economic regimes makes such averaged PDPs possibly misleading because they are aggregating period variations to obtain overall predictor to outcome relationships. However, in order to explore possible short-term patterns, we provide illustrative PDPs for the most recent 48-month window (2021-2024), in Appendix C. These plots provide for limited examination of feature relationships related to the more contemporary data while consciously characterizing them as non-stationary over the entire sample.

E. Conclusion

In this thesis I have systematically evaluated a wide range of machine learning models' predictive capability across a variety of testing frameworks, leading to a number of well-grounded conclusions about their possible utility in stock return forecasting. The in-sample results do imply that there is predictive power, however all methods relied on non-linear assumptions regarding the relationships in the data, with Random Forest achieving the best R^2 of 22.5% and XGBoost with 18.4%. Ultimately these values show that that the methods appear to be much better than the other linear models (Lasso: 2.88% , Ridge: 2.68%). However, in the broader out-of-sample predictions, the models appeared to struggle overall and suggested these methods are unsuitable in practical forecasting.

The expanding window results (Table 4.2) presented sharply to diminishing predictive power, which results show a decline in the r -squared to approaching zero (Random Forest: 0.785%, XGBoost: 0.318%). That pattern became more apparent under the stricter rolling window testing (Table 4.3) with Random Forest (the best performing model) providing an r^2 of only 0.196%. Based upon the degradation from in-sample performance, to out-of-sample performance and then from out-of-sample performance to rolling window results, it would appear that any predictive power these models may have exhibited in the in-sample results may have been a symptom of overfitting rather than predictive performance when looking at out-of-sample testing.

It is important to note that while R^2 values across the different tested framework, the error metrics of MAE and MSE did not vary to the same degree. Across all models the MAE ranged between 3.4-5.2%, regardless of the methodology or testing process used. Within this, indicates that all models' absolute predictive errors remain at an economically meaningful level. The MAE and MSE metrics consistency compared to the varying R^2 performance suggests that while different methods of machine learning are able to capture varying proportions of return variance, none are able to improve forecasting performance at a meaningful level.

The conclusions largely suggest that despite advances claims of modelling complex relationships better with machine learning methods, equity return prediction is fundamentally limited by inherent properties and knowledge of predictors (short-term market prices) and the wide range of assumptions used in machine learning techniques (non-linearity). While previous studies caution prospective predictive success in the out-of-sample context, especially in practical contexts such roll as rolling window forecasting, this suggests even these more complex techniques are unsuccessful at overcoming the noise and non-stationarity that exists within financial markets. It may be in future literature more studies could explore more novel model architectures and more nuanced hybrid approaches to account for these challenges, but in relation to the evidence in this body of work, these clearly warrant careful interpretation of apparent predictive success in similar, if not more, methodologies.

Appendix

A. Augmented Dickey-Fuller (ADF) Test Results

Variable	Dickey-Fuller Statistic	Lag Order	p-value
dp_cycle	-9.7772	10	< 0.01
ep	-3.6125	10	0.03148
de	-5.4640	10	< 0.01
CPI	-5.4948	10	< 0.01
GS10	-9.3720	10	< 0.01
tms	-4.9902	10	< 0.01
dfy	-4.4988	10	< 0.01
dfr_cycle	-9.1245	10	< 0.01
b_m	-4.9007	10	< 0.01
Ntis	-6.7009	10	< 0.01
b7ret	-7.7845	10	< 0.01
lagged_svar	-8.1890	10	< 0.01
stock_return	-10.2450	10	< 0.01

B. Tunings for Insample Machine Learning

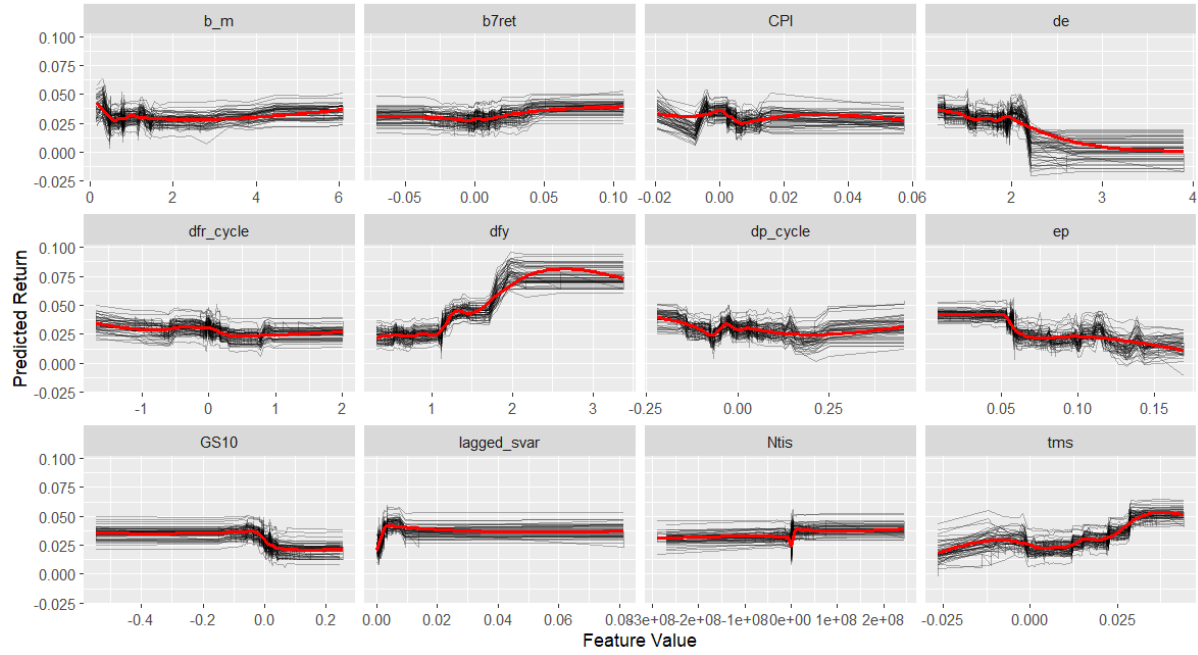
Algorithm	Parameter	Value	Description
XGBoost	objective	reg:squarederror	Regression with squared loss
	eta	0.05	Learning rate

Algorithm	Parameter	Value	Description
	max_depth	4	Maximum tree depth
	gamma	0.013	Minimum loss reduction for split
	subsample	0.2	Ratio of training instances subsampling
	colsample_bytree	0.8	Ratio of features subsampling per tree
	lambda	1	L2 regularization term
	alpha	0	L1 regularization term
	nrounds	100	Number of boosting rounds
	early_stopping_rounds	10	Stop if no improvement for N rounds
Random Forest	ntree	100	Number of trees in forest
	splitrule	extratrees	Splitting rule
	sampsize	0.2 * nrow(X)	Size of sample to draw
	replace	TRUE	Sample with replacement
	mtry	2	Number of variables randomly sampled
	maxnodes	11	Maximum number of terminal nodes
	nodesize	1	Minimum size of terminal nodes

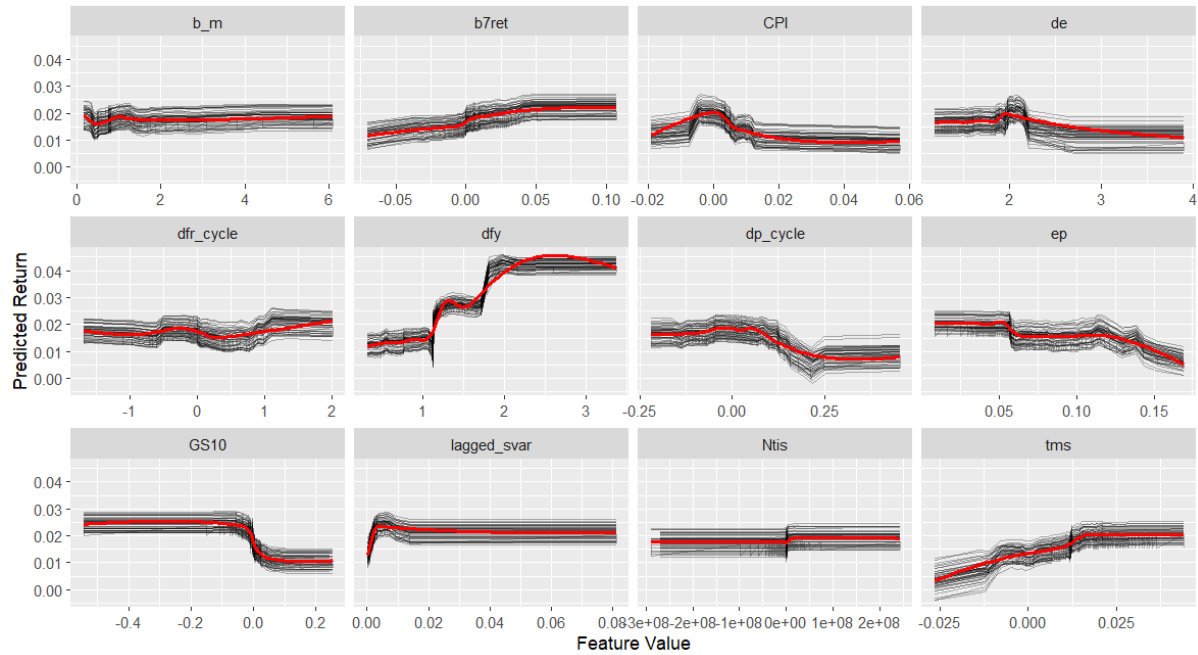
Table Appendix B Tunings for In sample predictions

C. PDP Plots

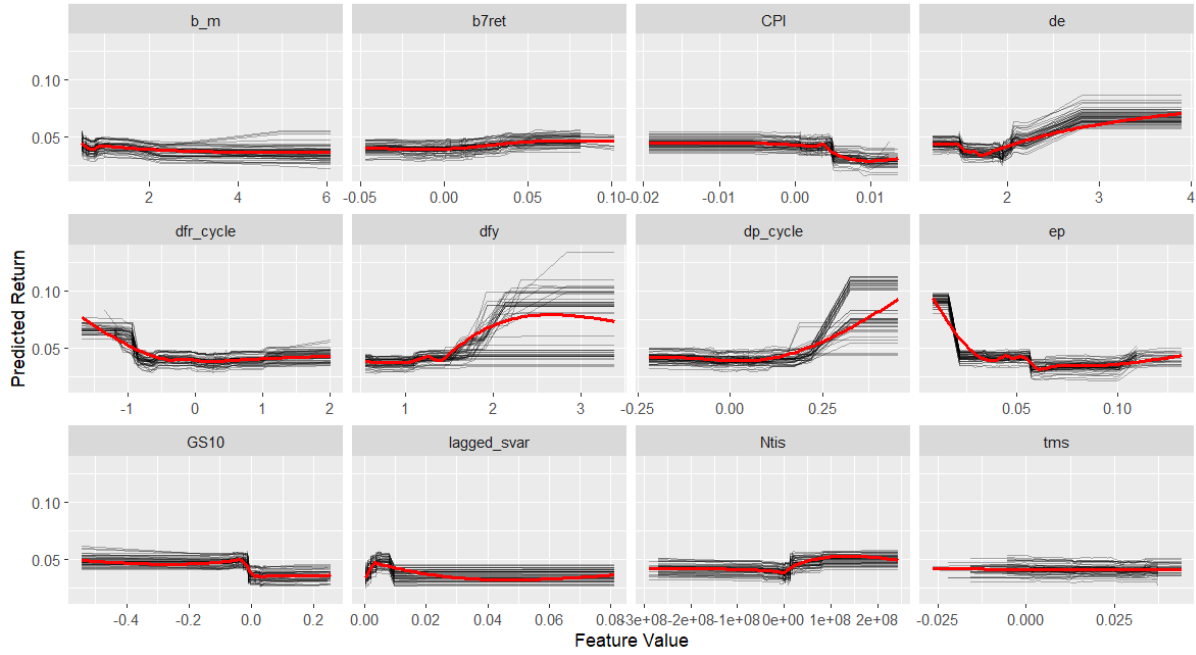
XGBoost Partial Dependence Expanding Window (Last 48 Months)



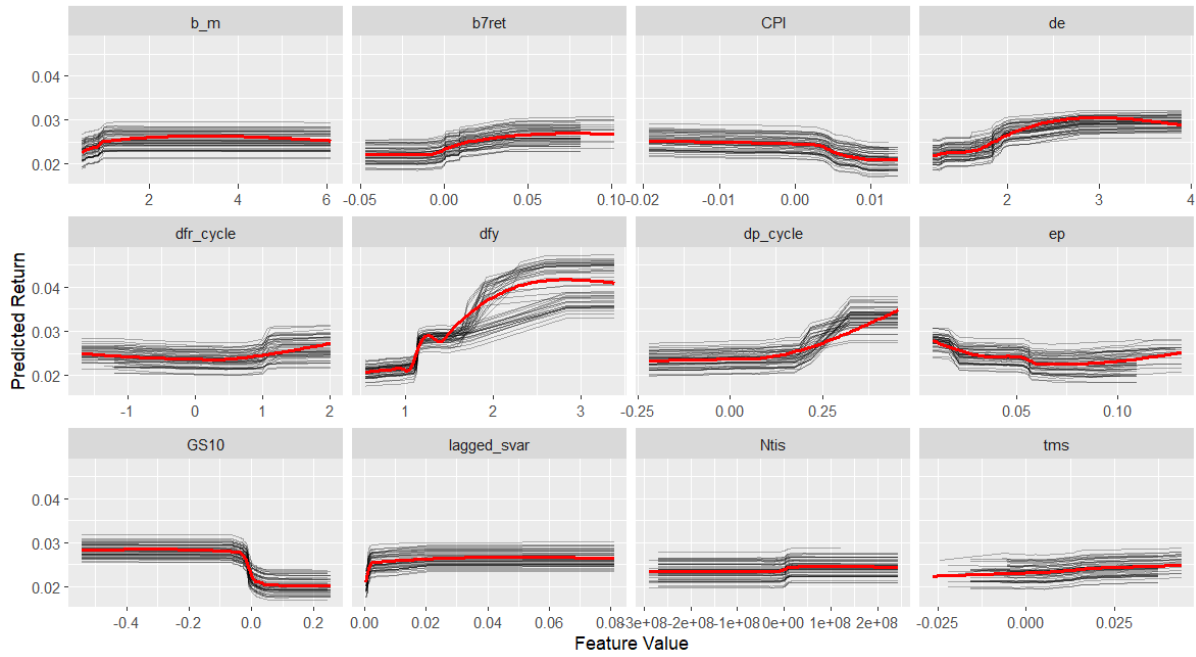
Random Forest Partial Dependence Expanding Window (Last 48 Months)



XGBoost Partial Dependence Rolling Window (Last 48 Months)



Random Forest Partial Dependence Rolling Window (Last 48 Months)



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